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Lithostratigraphy and Limestone Microfacies of the Jafnayn Formation (Paleocene to Early Eocene, Al-Khod, Sultanate of Oman): Deposition in a Restricted Lagoon with Intervals of Open Marine Conditions

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Abstract: We studied the Jafnayn Formation's lithostratigraphy, microfacies, depositional environment, and uncertain presence of the Paleocene/Eocene boundary and present the first detailed analysis of a 127-m-thick section using the standard microfacies (SMF)/facies zone (FZ) system. The formation is dominated by foraminiferal grainstones and packstones of SMF 18-FOR, followed by peloidal grainstones and packstones of SMF 16. Coral-red algae floatstones of SMF 8 occur sporadically. SMF 18-DASY appears only once. SMF 16 and 18-FOR suggest a restricted lagoon, whereas SMF 8 and 18-DASY reflect episodic open marine lagoonal conditions. The section consists of four lithostratigraphic units. Considerable detrital quartz near the base (Unit 1: 22%; sand, sandstone, conglomerate) and top (Unit 4: 40%; sand) coincides with a restricted lagoon affected by near-shore processes (base) and near-shore conditions (top). Open marine conditions show an elevated bioclast diversity in units 3 and 4. Unit 2 displays very thickly-bedded limestones of the restricted lagoon. The lagoon barriers likely consist of foraminifera and other allochems that were reworked from the lagoon to form shoals. Coral remains in units 3 and 4 suggest that coral build-ups in the upper part of the formation protected the lagoon as well. The early Eocene age of several basal *Alveolina* species in the lowermost 9 m of the studied section indicates that the section is incomplete, with the late Paleocene part and meters-thick basal yellow marl missing. The Paleocene/Eocene boundary is unexposed.

Keywords: near-shore environment; standard microfacies; foraminifera; quartz detritus; Rusayl Embayment

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1. Introduction

The Jafnayn Formation was introduced by Nolan et al. [1], who described the formation from the type section near Jafnayn (10 km SE of the Al-Khod section; Figure 1A,B), where it consists of 126-m-thick shallow marine limestones ([1], based on unpublished information by R. Crawford, no bibliographic details). The late Paleocene age of basal marly layers of the formation [2] has been confirmed by echinoids [1] and foraminifera [3–5]. The early Eocene age of the overlying limestones [2] was confirmed by foraminifera [3,4,6,7]. The lower part of the Jafnayn Formation is dominantly marly [4]. The thickness of the formation varies considerably [1]. In general, the Jafnayn Formation rests with a low-angle unconformity on the uppermost Cretaceous siliciclastic Al-Khod Formation

(Figure 2), while the upper contact appears to be conformable with the overlying Rusayl Formation [1] of the early Eocene (Figure 2). However, at the study site, the lower contact is faulted (see Section 2). Nolan et al. [1] subdivided the Jafnayn Formation into:

- 3 to 4 m of basal cream/buff, nodular crystalline limestone with ophiolite and chert clasts, laterally passing to yellow marl and limestone a few centimeters or meters thick with reworked clasts;
- 69 m of yellow marl and marly wackestone with one very thick weathering-resistant nodular, bioturbated echinoid-bearing bed of ~30 m thickness;
- 53 m of thickly bedded, cliff-forming beige/gray pack- and wackestones with grain- and rudstones towards the top, contains alveolinids, miliolids, rhodoliths, and corals with upward-increasing abundance.

The Jafnayn Formation was suggested to reflect deposition within a shallow shelf environment with low-energy conditions, except for the upper part of the formation with higher energy level conditions, as indicated by some coarse bioclastic debris [1]. The formation may have developed from a low-energy, normal marine, possibly lagoonal, shallow inner shelf (lower part) via a shallow, inner shelf, normal marine environment [4] to a shallow (less than 10 m), fairly high-energy environment with open marine shoals (upper part, [1]). The presence of coral patch reefs is assumed for the upper part [4].

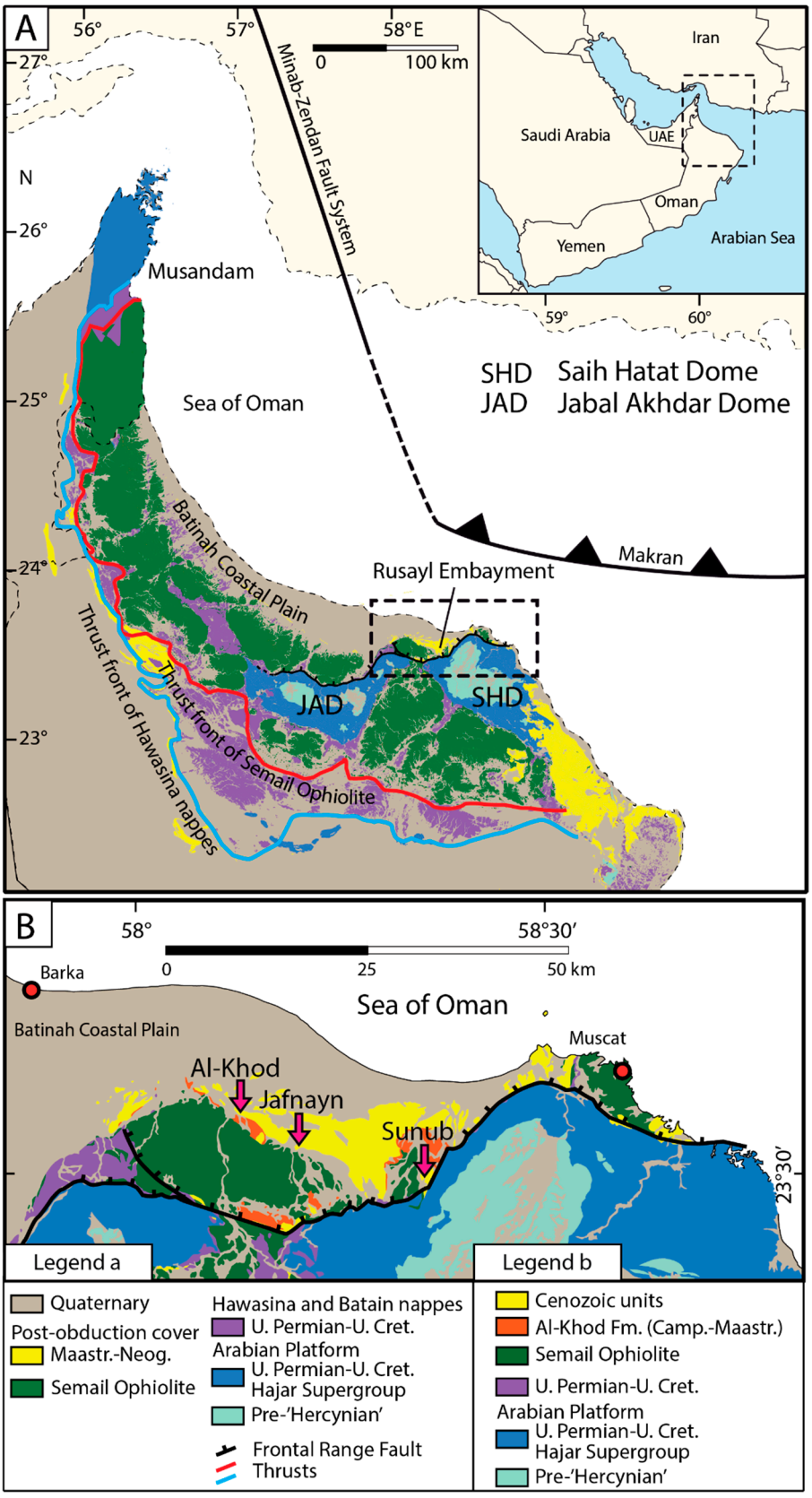


Figure 1. Location of the study area on geological maps. (**A**) Geology of the Oman Mountains. Box indicates map area shown in b. Modified after [8,9]. The Rusayl Embayment is the outcrop zone of the Al-Khod Formation and the Cenozoic formations north of the Jabal Akhdar and Saih Hatat domes. (**B**) Detailed map of the study site, located in the western part of the Rusayl Embayment,

after Béchennec et al. [8]. Frontal Range after [10]. The three locations marked with red arrows are locations in the Rusayl Embayment mentioned in the text.

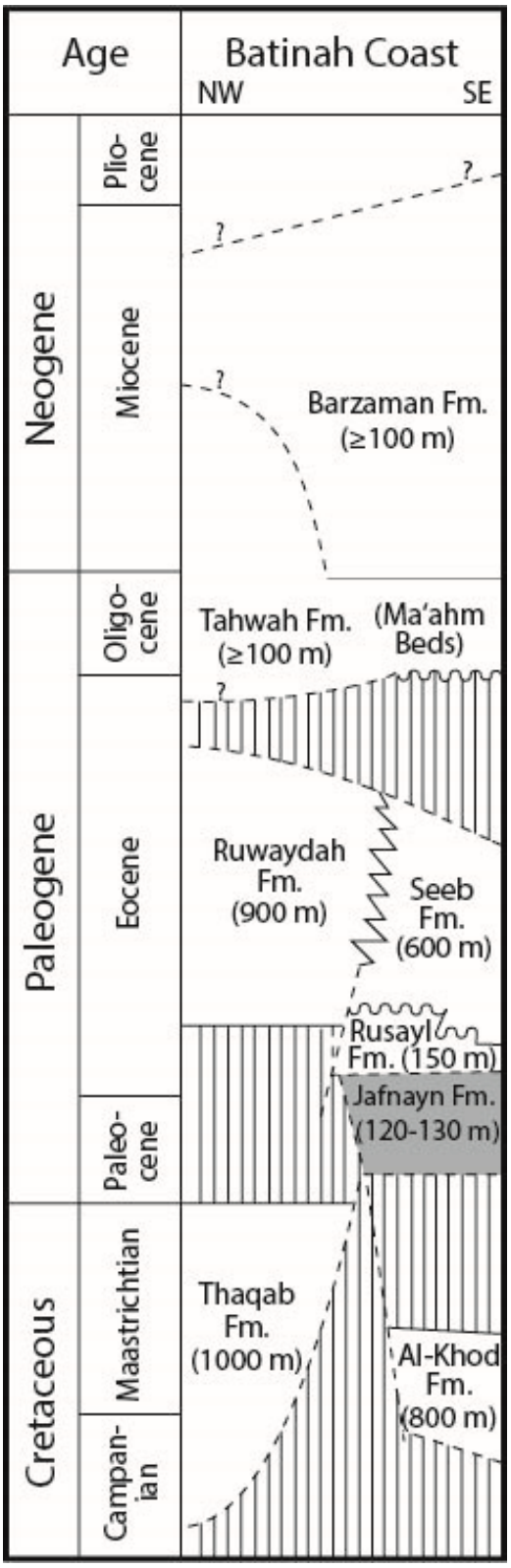


Figure 2. Stratigraphic overview of the Late Cretaceous and Cenozoic post-obductional formations of the study area, mainly after [1,11]. The thickness of the Rusayl Formation may vary greatly from region to region [1].

Dill et al. [12] studied the lithology, mineral composition, fossil content, and sedimentary structures of the uppermost 30 m of the Jafnayn Formation of the Al-Khod roadcut. They also determined the total carbon (TC), total organic carbon (TOC), and total sulfur contents (TS); analyzed carbon, oxygen, and sulfur isotopes; and suggested a depositional environment consistent with a shoreface/subtidal setting, including high-energy inter-island channels and tidal flats to a shallow lagoon with moderately reducing conditions. The tidal environment may include outer tidal flats [12].

Other studies focused on the Jafnayn Formation's foraminiferal content [3–6]. In particular, Serra-Kiel et al. [5] worked 170 km southeast of our study area, while Özcan et al. [6] focused on the western Rusayl Embayment, including the roadcut north of Al-Khod Village. Both studies confirmed a shallow marine environment based on the microfaunas.

Tomás et al. [7] paid special attention to the Jafnayn Formation's seagrass habitat and foraminiferal content in an area 150 km southeast of our study site and suggested a shallow marine low-energy platform environment within the photic zone, which was occasionally influenced by storm events, as well as the formation of high-energy shoals.

Clinoforms and microfacies in the lower part of the Jafnayn Formation in the eastern part of the Rusayl Embayment, near Sunub (Figure 1A,B), were studied by Mattern and Bernecker [13]. Based on the combined standard microfacies (SMF) systems of Wilson [14] and Flügel [15], they interpreted the lower part of the formation as lagoonal deposits.

Although the Jafnayn Formation in the studied outcrop has received some research attention (e.g., [6,12]), no detailed, bed-by-bed lithostratigraphic or microfacies analysis has ever been undertaken. Overall, the interpretations of the depositional environments at Al-Khod are vague, except for the uppermost 30 m [12].

The studied roadcut is a well-known and often visited geological focal point in Oman to conveniently study, compare, and sample the Cenozoic Jafnayn, Rusayl, and Seeb formations. Consequently, it has been the subject of significant published research [3,11,12,16–18]. However, it is uncertain whether the roadcut exposes rocks of the Paleocene Eocene Thermal Maximum.

Our work intends to fill the knowledge gaps by presenting a comprehensive lithostratigraphic and microfacies study of the Jafnayn Formation from the Al-Khod road section based on detailed logs, comprising information on the biota and detrital quartz contents as well as rock textures. This is aimed at robustly and objectively classifying the observed microfacies by applying the standard microfacies (SMF) and facies zone (FZ) system of Flügel [15]. By identifying foraminifera in thin-sections, we address the question of whether the Paleocene/Eocene boundary and, thus, the Paleocene Eocene Thermal Maximum (PETM) is located in the western roadcut section of this important outcrop. The results of our study are important for all future research related to the biostratigraphy, chemostratigraphy (e.g., solving paleoclimate questions), and regional paleofacies mapping of the period in question. Our results may rule in or rule out the famous Al-Khod roadcut for the consideration of such studies.

2. Geological Setting

The Late Cretaceous obduction of the Tethys-derived Semail Ophiolite and ocean floor sediments of the Hawasina Basin led to the formation of the Oman Mountains (e.g., [19–28]). Mountain building was associated with the development of two large domes, the Jabal Akhdar Dome in the West and the Saih Hatat Dome in the East [Figure 1A; review in 28]. The study area is located in the northern foothills of the Oman Mountains, in the Rusayl Embayment (Figure 1A,B), relatively close to the northeastern extremity of the Jabal Akhdar Dome (Figure 1A). At their northern flanks, the domes are accompanied by a major extensional fault, the Frontal Range Fault (Figure 1A,B), which was active from the Late Cretaceous to the early Eocene as well as from the Late Lutetian to the Oligocene due to orogenic collapse of the Oman Mountains [10,29]. The Rusayl Embayment is situated north of the Frontal Range Fault (Figure 1A,B).

In addition, the study area (Tertiary Ridge, Figure 3) belongs to the Hajar Wrench Corridor, which formed due to Oligocene/Early Miocene E/W-shortening related to oblique Arabia–India convergence [30]. As a result, the lower contact of the Jafnayn Formation at the Tertiary Ridge is generally marked by a fault [30,31], but this fault is not exposed at the SW end of the studied outcrop as it is covered by scree. Therefore, it is uncertain how much of the lower part of the Jafnayn Formation is missing at the base of the studied section.



Figure 3. Superb exposure of the Jafnayn Formation at the studied western roadcut north of Al-Khod Village. The dip is to the NE. This outcrop is part of a hogback known as the Tertiary Ridge [31].

Post-obductional deposition started with the fluvial and coastal Late Campanian?/Maastrichtian? Al-Khod Formation and its lateral equivalent, the Qahlah Formation, nonconformably on the ophiolite [1,32,33]. The Al-Khod Formation is characterized by conglomerate, sandstone, and shale and may measure 800 to 860 m in thickness [1,33]. With the overlying Jafnayn, Rusayl, and Seeb formations as well as the Ma'ahm Beds and Barzaman Formation, these post-obductional formations are dominated by shallow marine limestones of Paleocene to Miocene (Pliocene?) age (Figure 2) [1,3–7,11–13,17,18,34–40].

Post-obductional deposition ensued under generally stable tectonic conditions [41], and subsidence was slow and of uncertain origin [42]. The Jabal Akhdar and Saih Hatat domes experienced differential exhumation and doming during the Cenozoic [43], forming the main structures of the Oman Mountains in the region of the study area (Figure 1).

During the Paleocene and Eocene, the study area's paleogeographic position was between the Tropic of Cancer and the equator [44]. In this tropical setting, shallow marine carbonate sediments accumulated on either side and in close proximity to the Oman Mountains/domes during the Paleocene and Eocene, including the Jafnayn Formation [1, their Figure 12].

The long-term eustatic sea-level curve indicates a moderate sea-level rise during the Paleocene, followed by a drop near the Paleocene/Eocene transition with no further perturbation for the remainder of the early Eocene [45].

During the Paleocene, the global average temperatures overall increased [46], climaxing in the Paleocene–Eocene Thermal Maximum (PETM) at the Paleocene/Eocene boundary [47], which triggered the widespread Larger Foraminiferal Turnover (LFT) described by Scheibner et al. [48]. The Eocene represents the warmest epoch of the Cenozoic era [49]. According to quantitative comparison with fully coupled climate model simulations, global average temperatures were about 29, 26, 23, and 19 °C during the early, early middle, late middle, and late Eocene, respectively [49]. It is questionable whether the Jafnayn Formation at the studied outcrop includes the PETM event.

3. Materials and Methods

We studied an excellent outcrop in which the Jafnayn Formation is continuously exposed (Figure 3). Following a detailed analysis in the field, four stratigraphic logs (referred to as "log segments 1 to 4") were measured, and 48 thin sections were prepared. Each log is covered by 12 thin sections (one sample every ~2.6 m), which were analyzed for faunal content, non-skeletal grains, and limestone textures to determine the SMFs and FZs *sensu* Flügel [15], recently tabulated for practicality [50], to constrain the depositional setting. The SMF/FZ system is based on skeletal and non-skeletal grain data, the frequency of such grains as well as limestone textures and sedimentary structures [15]. The use of the SMF/FZ system kept the following facies descriptions and interpretations brief and concise, emphasizing the lithology and skeletal/non-skeletal components. In general, not all limestones meet SMF definitions ([15], p. 712).

While the upper contact of the Jafnayn Formation with the Rusayl Formation is well exposed (Figure 3), the lower contact with the Al-Khod Formation is covered by scree (Figure 3). Moreover, large-scale structural work along the Tertiary Ridge (Figure 3) north of Al-Khod showed that a fault exists between the Al-Khod Formation and the Jafnayn Formation at the SW-limit of the Tertiary Ridge [31]. Thus, we paid special attention to the foraminifera of the oldest exposed beds of the Jafnayn Formation to reach our stratigraphic goal. Foraminifera were identified in standard thin-sections. For identification and age assignments, we compared our findings with those of benchmark publications (e.g., [5]) and other relevant literature.

The terms "Log Segment 1" to "Log Segment 4" denote stratigraphic sections that are thin enough to fit onto one graphic illustration, allowing for all essential details to be depictable and perceivable. The terms "Unit 1" to "Unit 4" refer to lithostratigraphically distinctive parts of the section. Log segments and units do not correlate.

4. Results

We investigated the Jafnayn Formation along the road that leads from the north into Al-Khod Village (Figures 1 and 3; west side of road; coordinates 23°34′18.74″ N/58° 7′40.11″ E) and cuts through a hogback with the Jafnayn Formation at its base (Figure 3). The thickness of the exposed part of the Jafnayn Formation in this roadcut measures 127 m. The observed detrital quartz grains are either monocrystalline or polycrystalline. In the latter case, this is due to a metamorphic or chert origin.

4.1. Log Segment 1

The contact between the underlying Al-Khod Formation and the base of the Jafnayn Formation is not exposed (Figure 3). The basal parts of the Jafnayn Formation contain thin to medium marl layers (Figure 4) and limestone beds with reworked chert clasts (conglomerates; Figures 4 and 5). In the lower 39.5 m, the succession evolves upward from thick-bedded packstones and grainstones to very thick-bedded grainstones. Gastropods and common larger benthic foraminifera (LBF) can be seen with the naked eye. Alveolinids and nummulitids are common. Bioturbation is minor (Figure 4). Cross-bedding, convolute lamination, and gutter casts (Figure 6A) occur only once. Thin conglomerate horizons occur within the limestone beds (Figure 4) and are matrix-supported with

subrounded, pebble-sized clasts of quartz and chert (at ~5 m, ~12.5 m and 21 m above base; Figure 4). Two thin sandstone horizons comprising angular, medium-to-very-coarse quartz sand occur at ~6 m above the base of the section (Figure 4).

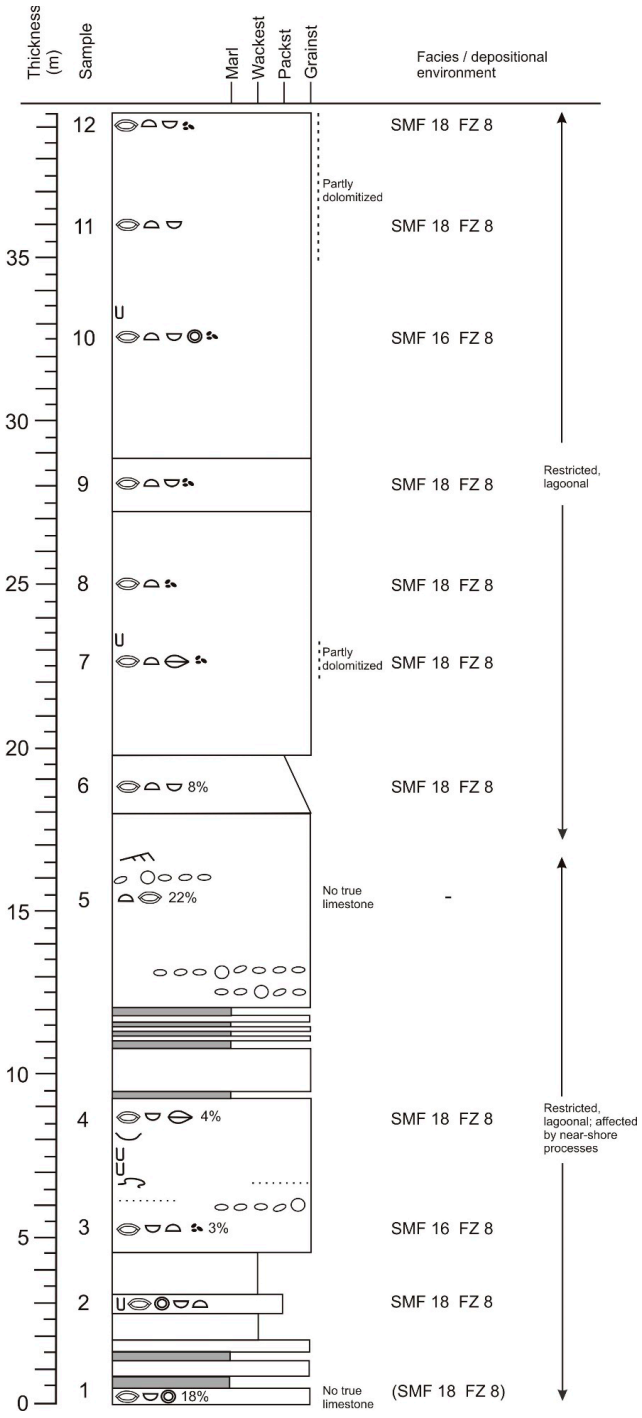


Figure 4. Detailed lithostratigraphic and facies log of Log Segment 1 based on field and microscopic observations. The frequency of the depicted skeletal grains decreases from left to right (note order!). Also, the frequency of the depicted non-skeletal grains decreases in the same direction (note order!). Percentages relate to detrital quartz content. For legend, see Figure 5.

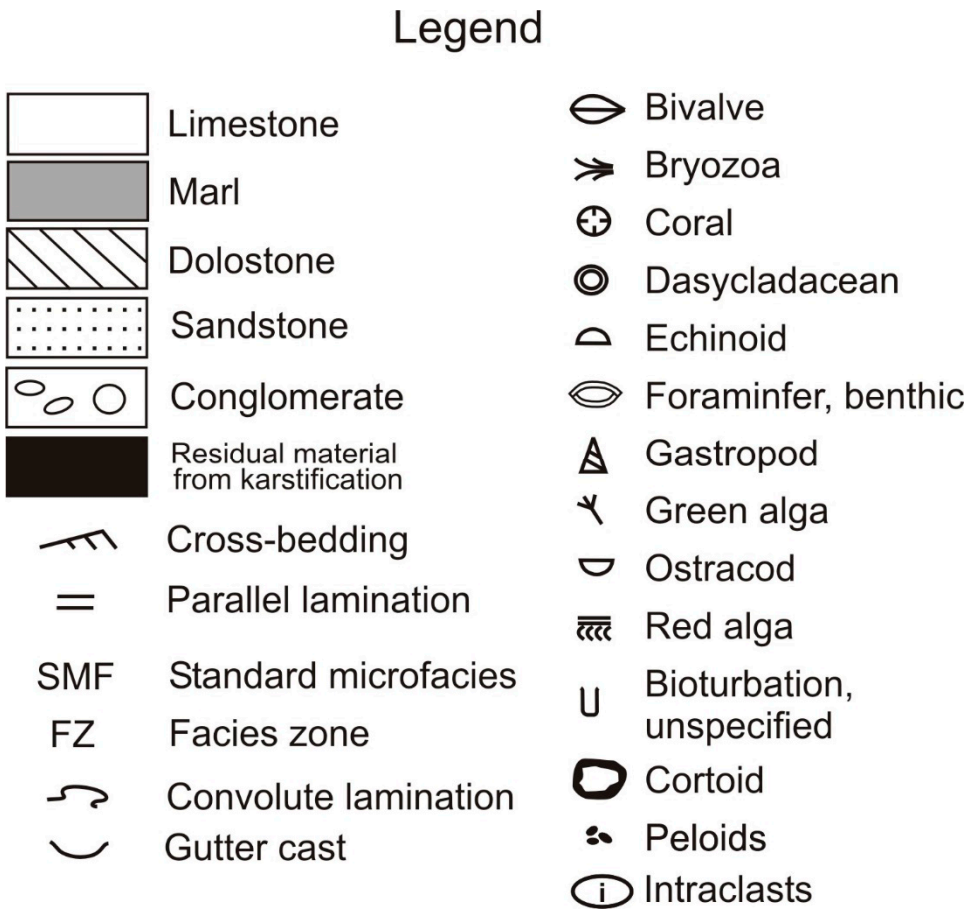


Figure 5. Legend for the logs of Figures 4, 7, 9, and 11. No true limestones are limestones with >10% of detrital quartz.

Thin-section analyses are summarized in Table A1 of Appendix A with identified foraminiferan taxa. Most thin-sections display foraminiferal grainstone textures (Figure 6B) prevailing over finer grained peloidal grainstones (Figure 6C). Detrital quartz occurs sporadically in the lower 22 m (Figures 4 and 6D). Overall, benthic foraminifera are the most common bioclasts, followed by echinoids. Less common are dasycladacean green algae (Figure 6E) and smooth-shelled ostracods (Figure 6E). Bivalves are rare. Although gastropods were observed with the naked eye in the field, they were not found in thin-sections. Foraminifera can be sizeable, often reaching 3–4 mm in diameter, and a few echinoid fragments measure several mm in length. Miliolids also occur as "small" foraminifera (<0.5 mm; [15], p. 458). Non-skeletal grains (peloids and cortoids) generally occur in limited numbers, except in grainstone textures, where they dominate. Peloids are much more common in the upper part.

Among the twelve samples, eight represent SMF 18-FOR (bioclastic grains- and packstone with abundant and rock-building foraminifera [15]), while one is similar to SMF 18-FOR, but also contains detrital quartz of >10%. Two samples are assigned to SMF 16-NON-LAMINATED (non-laminated peloid grain- and packstone [15]). One sample does not match an SMF. Dolomitization was observed in the middle and upper parts of Log Segment 1 (Figure 4) in samples 7, 11, and 12 (Figure 6F). The dolomite crystals are mostly non-mimic, limpid rhombs of ≤0.1 mm. In addition, there are non-mimic, rhombic crystals with dark cores and limpid rims (≤0.15 mm; Table A1). Bioclasts that survived dolomitization include echinoids, foraminifera with porcelaneous tests (miliolids), and some smooth-shelled ostracod valves.

In summary, the lower part of this succession is characterized by a few thin-to-medium marl layers, detrital quartz, and low faunal diversity, while the upper part displays

more thickly bedded, cleaner carbonates with higher faunal diversity (Figure 4). SMF 18-FOR is the dominant microfacies.

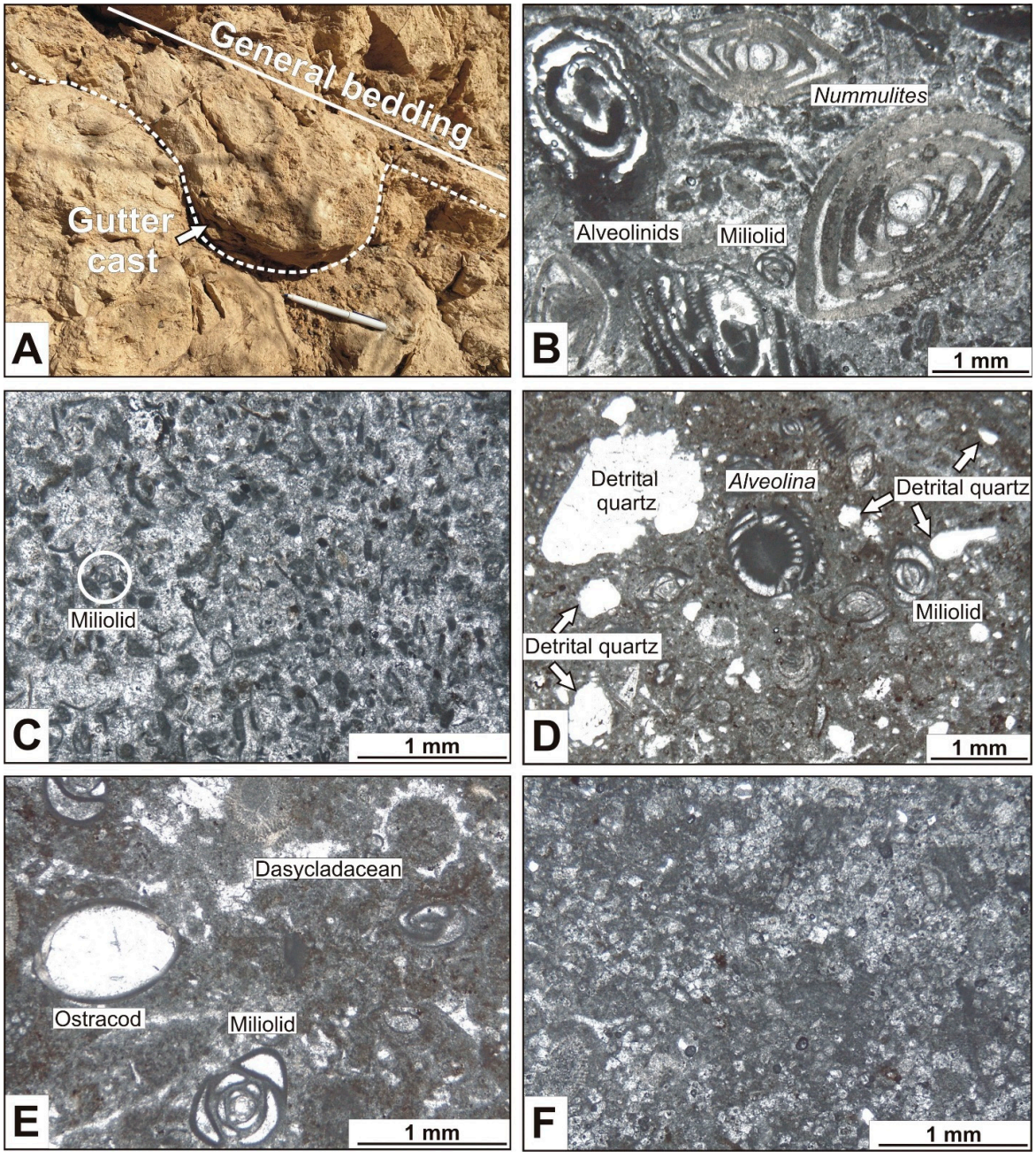


Figure 6. Field photograph (**A**) and thin-section photomicrographs (B-F) of Log Segment 1. (**A**) Gutter cast, oriented NNW/SSE. (**B**) Foraminiferal grainstone with benthic foraminifera, including *Nummulites* sp. (not *Nummulites globulus*!) and alveolinids, representing SMF 18-FOR. Sample 8, PPL. (**C**) Fine-grained peloidal grainstone, representing SMF 16-NON-LAMINATED. Sample 3, PPL. (**D**) Impure foraminiferal limestone with benthic foraminifera, including *Alveolina* sp. and miliolids, and detrital quartz from the base of the formation. Sample 1, PPL. (**E**) Dasycladacean alga, ostracod, and benthic foraminifera (mostly miliolids) of a foraminiferal packstone, representing SMF 18-FOR. Sample 2, PPL. (**F**) Local dolomitization of a foraminiferal grainstone. Note minute dolomite rhombohedra! Sample, 11, PPL.

4.2. Log Segment 2

The overlying 32.5 m (39.5 to 72 m above base of the road outcrop) are characterized by very thick-bedded pack- and grainstones (Figure 7). This log segment does not show

sedimentary structures. According to field observations, conglomerates, sandstones, and detrital quartz are absent. Foraminifera, red algae, and gastropods can be observed with the naked eye. A dolostone bed is located at 61 m above base (Sample 7, Figure 7).

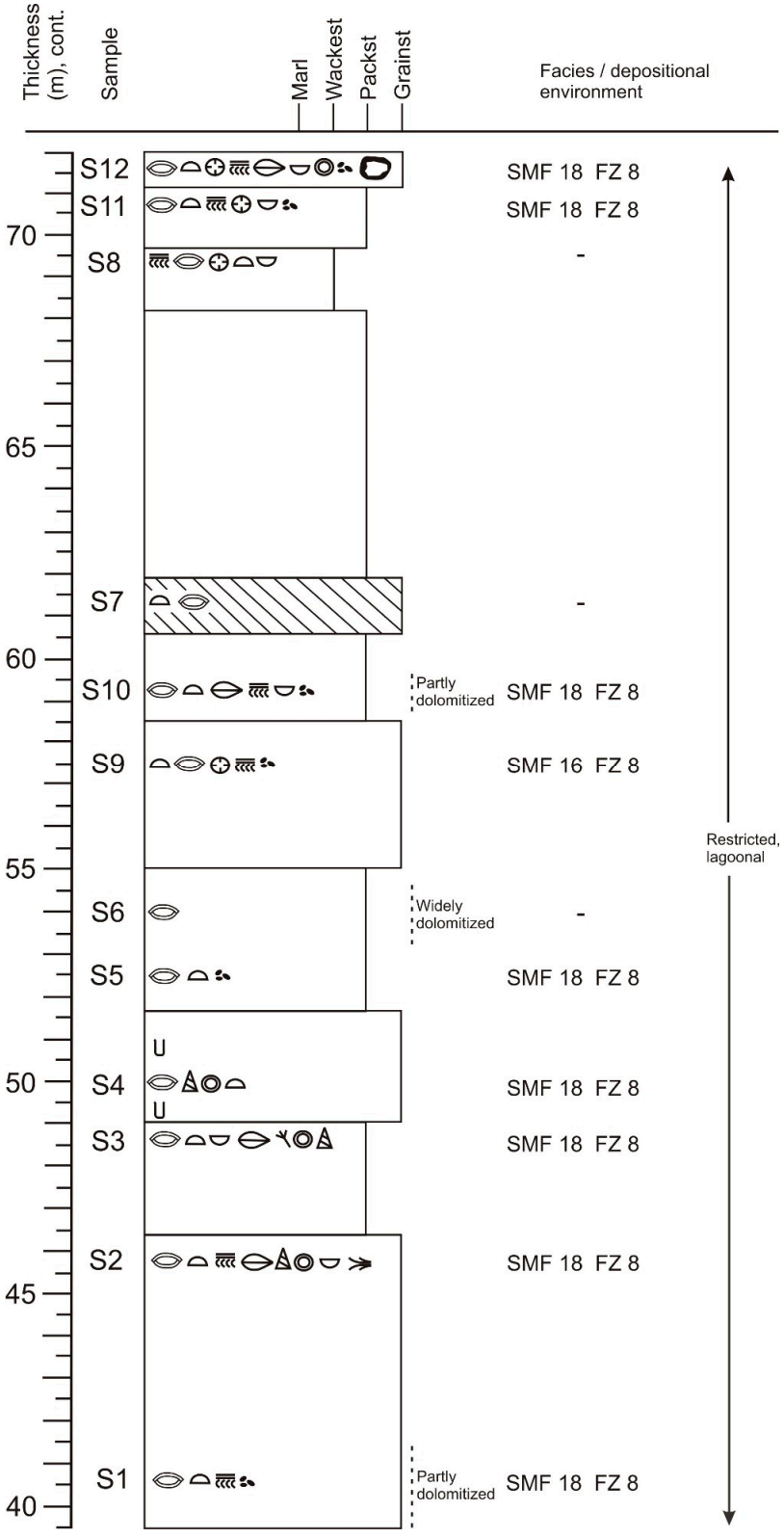


Figure 7. Detailed lithostratigraphic and facies log of Log Segment 2 based on field and microscopic data. The frequency of the portrayed skeletal grains decreases from left to right (note order!). Also, the frequency of the depicted non-skeletal grains decreases in the same direction (note order!). Percentages relate to detrital quartz content. For the legend, see Figure 5.

The thin-section analyses are summarized in Table A2 of Appendix B with identified foraminiferal taxa. Detrital quartz is absent and partial dolomitization was observed at the bottom and in the middle part of the segment. The dolomite crystals are predominantly non-mimic rhombs with dark cores and limpid rims (either ≤0.1 mm or ≤0.15 mm). The remainder are non-mimic (Table A2). In the dolostone, at 61 m, a few relict echinoid fragments and porcelaneous foraminiferal tests are preserved. The dolomite crystals in the dolostone are mostly dark, anhedral, and non-mimic (≤0.2 mm; Table A2). Benthic foraminifera and echinoids (Figure 8A) are abundant. Foraminifera are sizeable. Foraminifera are the predominant bioclast type, although three samples are dominated by echinoids or red algae. At 57.5 m above base, corals (Figure 8B) are present, with poritid corals being more common than non-poritid scleractinians. Corals are frequent in the uppermost 3 m of Log Segment 2 (Figure 7) and are associated with red algae (Figure 8B–D), which sometimes encrusted the corals (Figure 8B). Bivalves, smooth-shelled ostracods, dasycladacean algae, and gastropods occur subordinately. We also noticed a bryozoan-like encrustation (encrusting foraminifer?) structure (Figure 8E). Peloids and cortoids occur more frequently than in Log Segment 1. Among the twelve samples, eight represent SMF 18-FOR and one SMF 16-NON-LAMINATED. Three samples could not be assigned to an SMF.

In summary, Log Segment 2 consists of pure carbonates, with poritid corals and red algae becoming significant in the uppermost part (Figure 7). The standard microfacies are dominated by SMF 18-FOR (Figure 7).

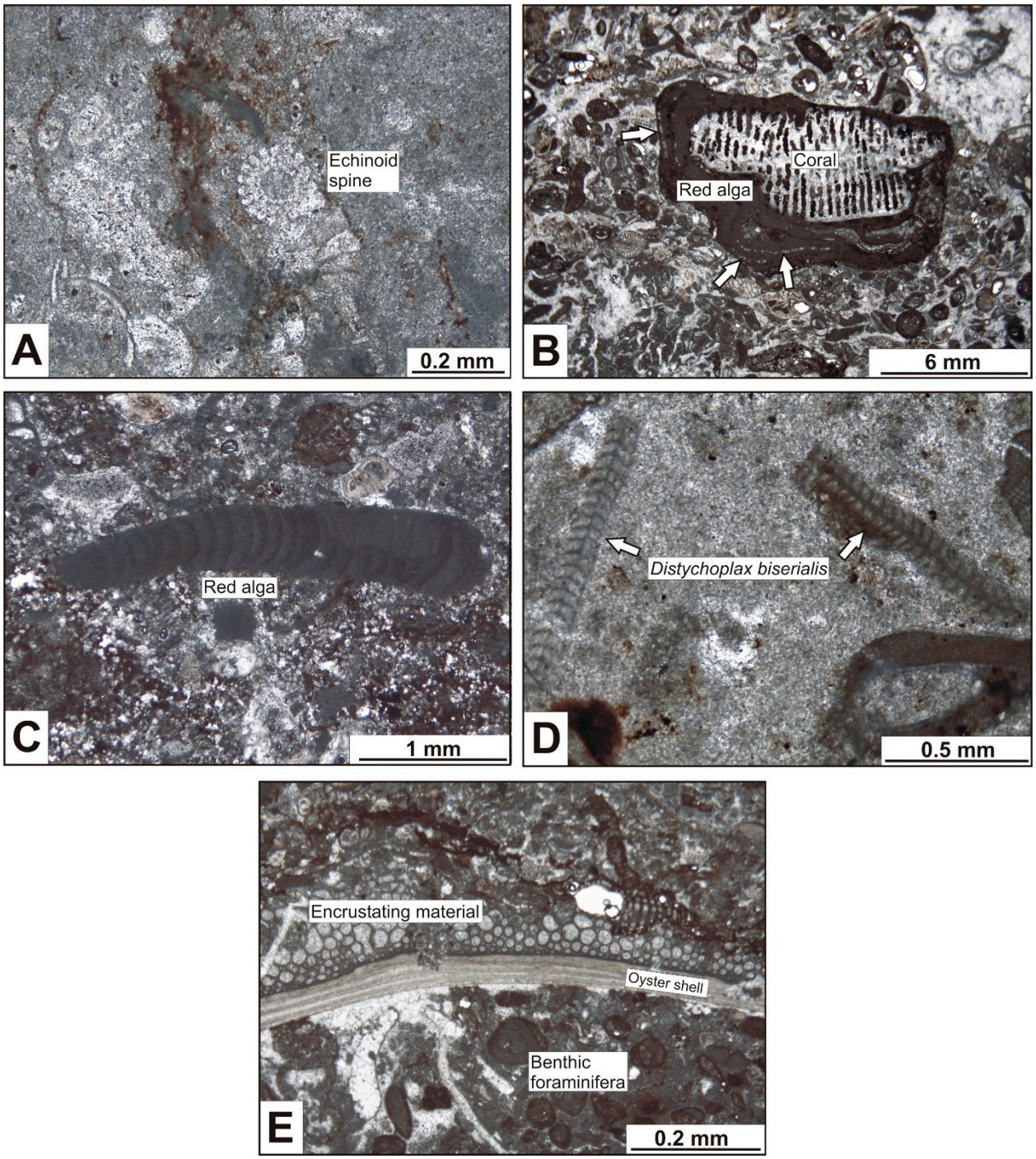


Figure 8. Thin-section photomicrographs of Log Segment 2. (**A**) Transverse section of an echinoid spine in a local packstone (no foraminifera shown) of an overall foraminiferal packstone, representing SMF 18-FOR. Dark lines may be pressure solution seams. Sample S3, PPL. (**B**) Coral with septae in longitudinal section. Spaces between the septae are filled with micrite. Coral is completely encrusted by red alga, which exhibits sporangia (arrows). Foraminiferal grainstone representing SMF 18-FOR. Sample S12, PPL. (**C**) Twig-like, rhodophytic red alga in a wackestone. Sample S8, PPL. (**D**) *Distychoplax biserialis* in a local "wackestone" of an overall foraminiferal packstone, representing SMF 18-FOR. Sample S2, PPL. (**E**) Oyster shell with bryozoan-like encrustation?/encrusting foraminifer? on the convex side of the oyster shell. Numerous benthic foraminifera are on the concave side of the shell. Foraminiferal packstone representing SMF 18-FOR. Sample S10, PPL.

4.3. Log Segment 3

This log segment extends from 72 to 107 m above the base of the studied section and is dominated by thick-bedded and very thick-bedded limestones (Figure 9). Fossils that can be seen in the field are gastropods, foraminifera, and red algae. Light bioturbation is common.

Thin-section analyses are summarized in Table A3 of Appendix C with identified foraminiferal taxa. Coral-red algae floatstones (Figure 10A) occur for the first time, while the basal and middle parts are dolomitized/partly dolomitized. There are different dolomite crystals of similar size (≤0.15 mm) and are either limpid, non-mimic rhombs; beige, anhedral, and non-mimic crystals; or non-mimic rhombs with dark cores and limpid rims (Table A3). The lower part of this segment is dominated by grainstones while the textures of the upper part are more diverse (fining-up trend). Benthic foraminifera continue to be the most common bioclast type but are of smaller size than those seen in log segments 1 and 2. Three samples are dominated by red algae, corals or dasycladacean algae. Poritid corals (Figure 10B) and red algae (Figure 10C) are especially abundant between 77 and 79 m above the formation base, and dasycladaceans (Figure 10D) are relatively common between 83 and 86 m. Red algae may encrust the corals (for corallinacean red alga, see Figure 10C), especially in the floatstones at 79 and 89.5 m (Figure 10B,C). Echinoids are rarer than in log segments 1 and 2, and green algae, bivalves, smooth-shelled ostracods, and gastropods are rare. Sample "3." includes a sizeable corallinacean red alga. (Figure 10E). Overall, peloids and cortoids are most frequent in this segment, and there are occasional micritic, miliolid-bearing intraclasts (Figure 10B). A minor influx of detrital quartz was observed at 76, 87, and 106 m above base. Four samples are assigned to SMF 16-NON-LAMINATED, three to SMF18-FOR, two to SMF 8 (wacke- and floatstones with whole fossils and well-preserved endo- and epibiota [15]), and one to SMF 18-DASY (bioclastic grains- and packstone with abundant and rock-building dasycladaceans [15]), while two samples could not be assigned to an SMF.

In summary, Log Segment 3 displays more lithological and microfacies diversity than log segments 1 and 2. The floatstones of SMF 8 are highlighted.

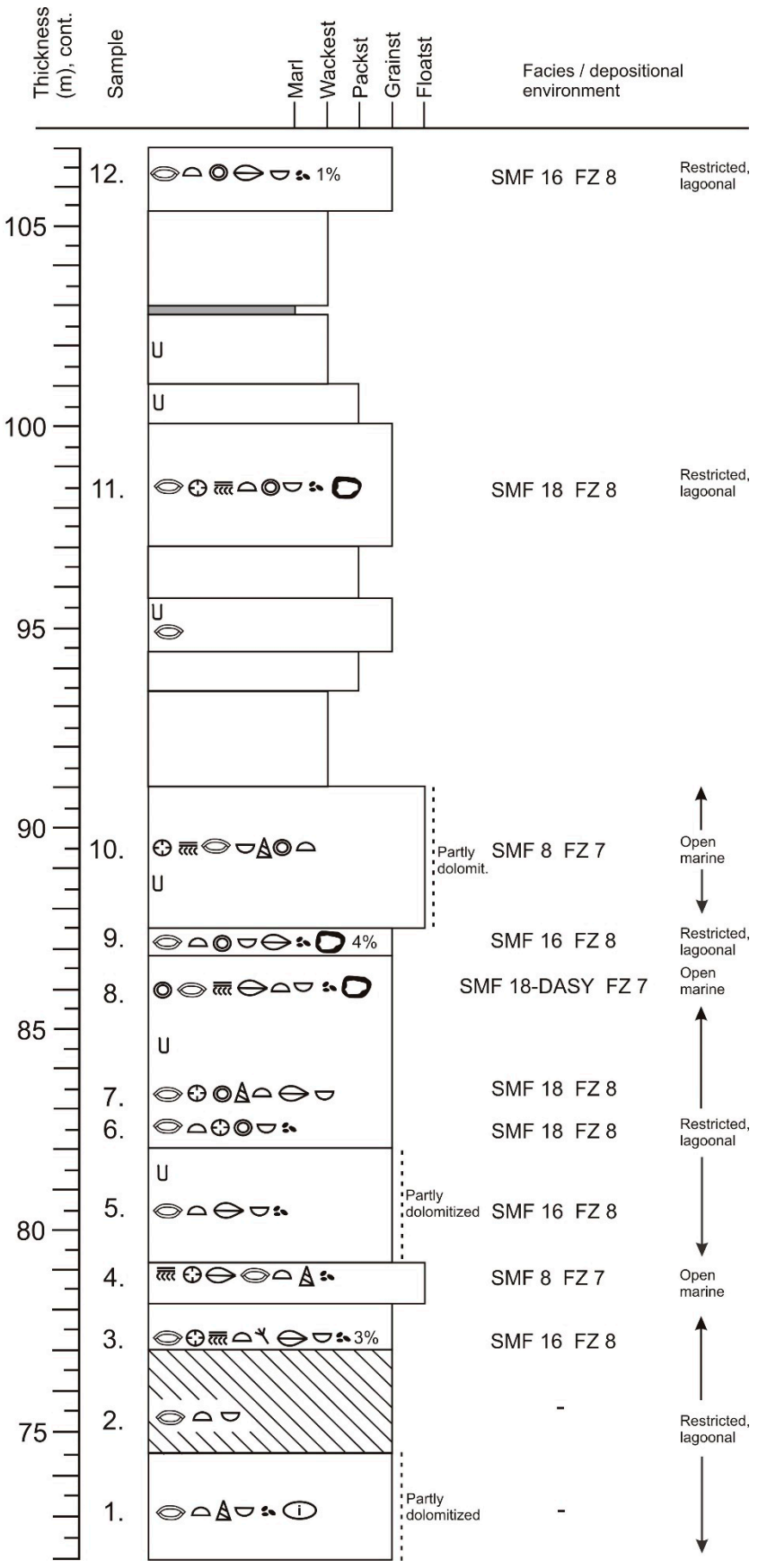


Figure 9. Detailed lithostratigraphic and facies log of Log Segment 3 based on field and microscopic data. The frequency of the depicted skeletal grains decreases from left to right (note order!). Also, the frequency of the depicted non-skeletal grains decreases in the same direction (note order!).

Percentages relate to detrital quartz content. Please note that dots after the numbers are part of the sample code to avoid confusion with the sample codes of Log Segment 1. For the legend, see Figure 5.

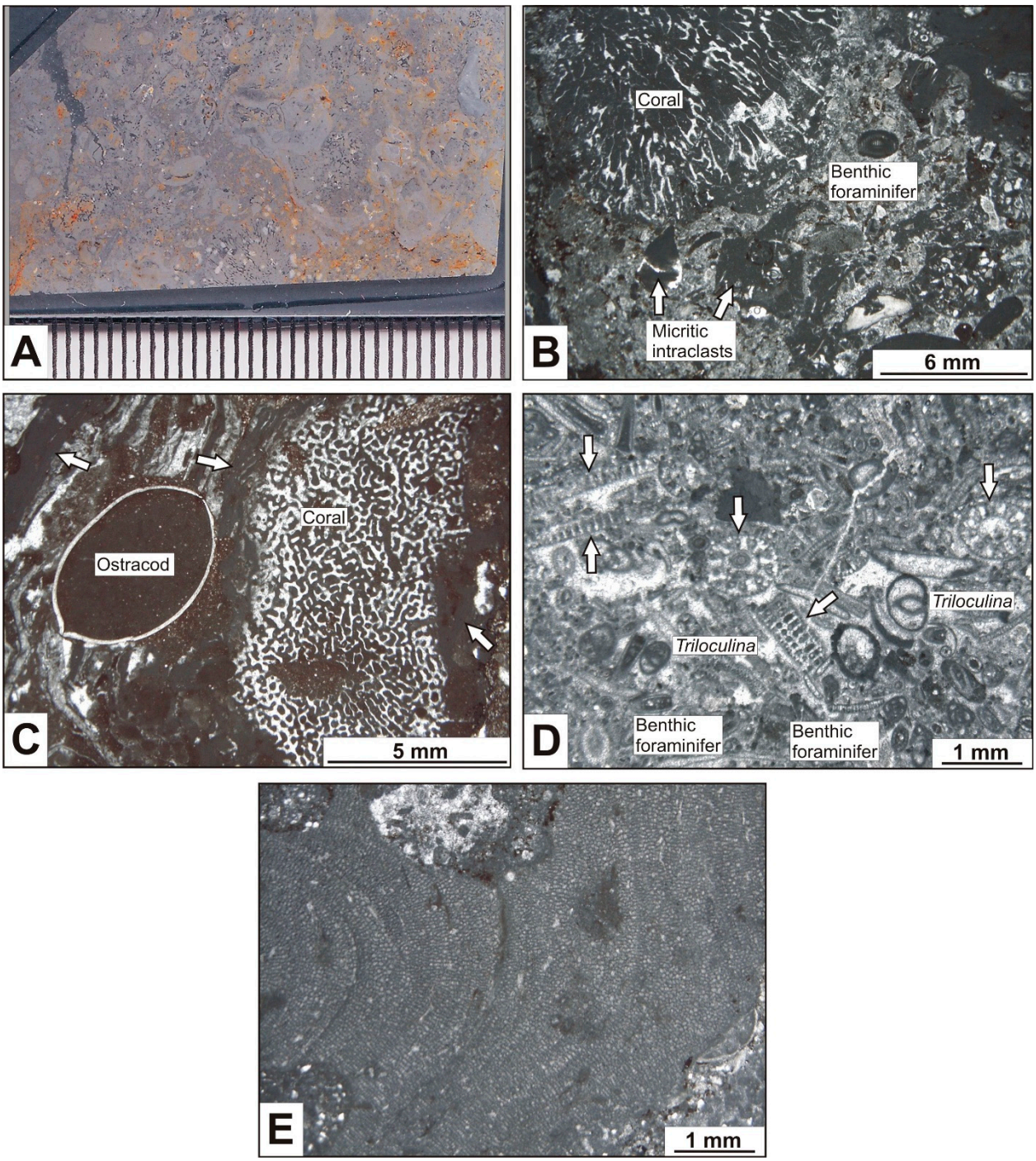


Figure 10. Digital scan of thin-section (**A**) and thin-section photomicrographs (**B**–**E**) of samples from Log Segment 3. (**A**) Digital scan of a thin-section of a coral–red algae floatstone allowing observation of the floatstone texture. The sample represents SMF 8. A black background was used to optimize the contrast to demonstrate the floatstone texture. The ruler at the bottom has millimeter spacing. Sample "4.". (**B**) Poritid coral and micritic intraclasts in coral–algae floatstone next to benthic foraminifer, representing SMF 8. Dasycladacean grainstone, Sample "4.", PPL. (**C**) Red algae (arrows), encrusting poritid coral, representing SMF 8. Sample "10.", PPL. (**D**) Dasycladacean (group *Thyrsoporelleae*) grainstone with benthic foraminifera, including *Triloculina* sp., representing SMF18-DASY. Dasycladaceans are marked by arrows. Sample "8.", PPL. (**E**) Corallinacean red alga. Peloidal grainstone, representing SMF 16. Sample "3.", PPL.

4.4. Log Segment 4

This log segment extends from 107 to 127 m above the base of the logged section and is dominated by thick-bedded and very thick-bedded limestones (Figures 11 and 12A). Corals, bivalves, ostracods, and, occasionally, sizeable LBF (Figure 12B) can be seen in the field. Sedimentary structures are rare, with parallel lamination observed in a few places. Very coarse quartz sand including a few quartz granules was observed in the upper few meters of the formation (Figure 11). Five "horizons" of reddish/brownish residual material from karstification were recognized (Figures 11 and 12C,D). They may display parallel lamination and seem to broadly follow the general bedding, although some thickness changes are recognized as well as dissolution features that are clearly non-parallel to bedding (Figure 12C,D). Field evidence did not provide any clues as to the type of karstification (syn- or post-depositional). In the field, detrital quartz grains were observed in the top ~10 m. The contact to the overlying Rusayl Formation is well-exposed (Figure 12A). It is marked by an abrupt depositional change from limestone (Jafnayn Formation) to marl (Rusayl Formation) (Figure 12A).

The thin-section analyses are summarized in Table A4 of Appendix D with identified foraminiferal taxa. Partly dolomitized limestones occur at ~109 m, 112 m, ~116 m, and 126 m above formation base (Figure 11). The dolomite crystals are mostly beige, anhedral, and non-mimic, measuring ≤0.2 mm. Benthic foraminifera are the dominating bioclast type but are locally exceeded by corals (Figure 13A), red algae, echinoids, dasycladaceans (Figure 13B), and bivalves (Figure 13C). The size of the foraminifera can exceed 1 cm (Figure 12B). Bivalves, ostracods (smooth-shelled), and gastropods are rare. Six of the twelve samples match an SMF, with three assigned to SMF 18-FOR, two to SMF 16, and one to SMF 8. Limestones, rich in quartz detritus (20–30%), characterize the uppermost 6 m of the formation (Figures 11 and 13C).

In summary, SMF 8 (floatstone) recurs in the uppermost 20 m (Figure 11). The absence of SMF types in the uppermost 6 m coincides with a significant increase in quartz detritus (no proper limestones, Figure 11).

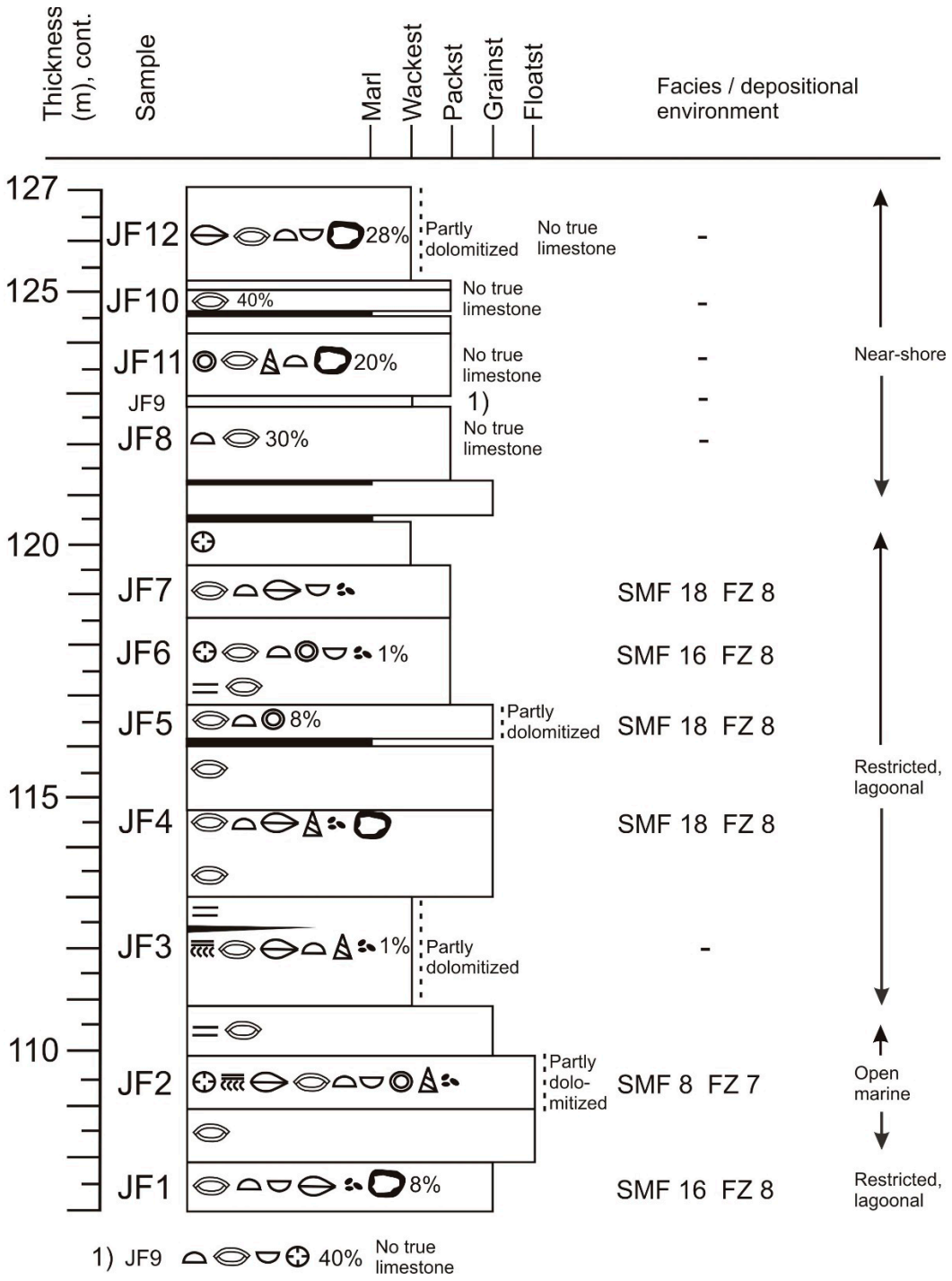


Figure 11. Detailed lithostratigraphic and facies analysis of Log Segment 4 based on field and microscopic data. The frequency of the portrayed skeletal grains decreases from left to right (note order!). Also, the frequency of the depicted non-skeletal grains decreases in the same direction (note order!). Percentages relate to detrital quartz content. The three accumulations of the residual material from karstification are parallel or roughly parallel to bedding. For the legend, see Figure 5.

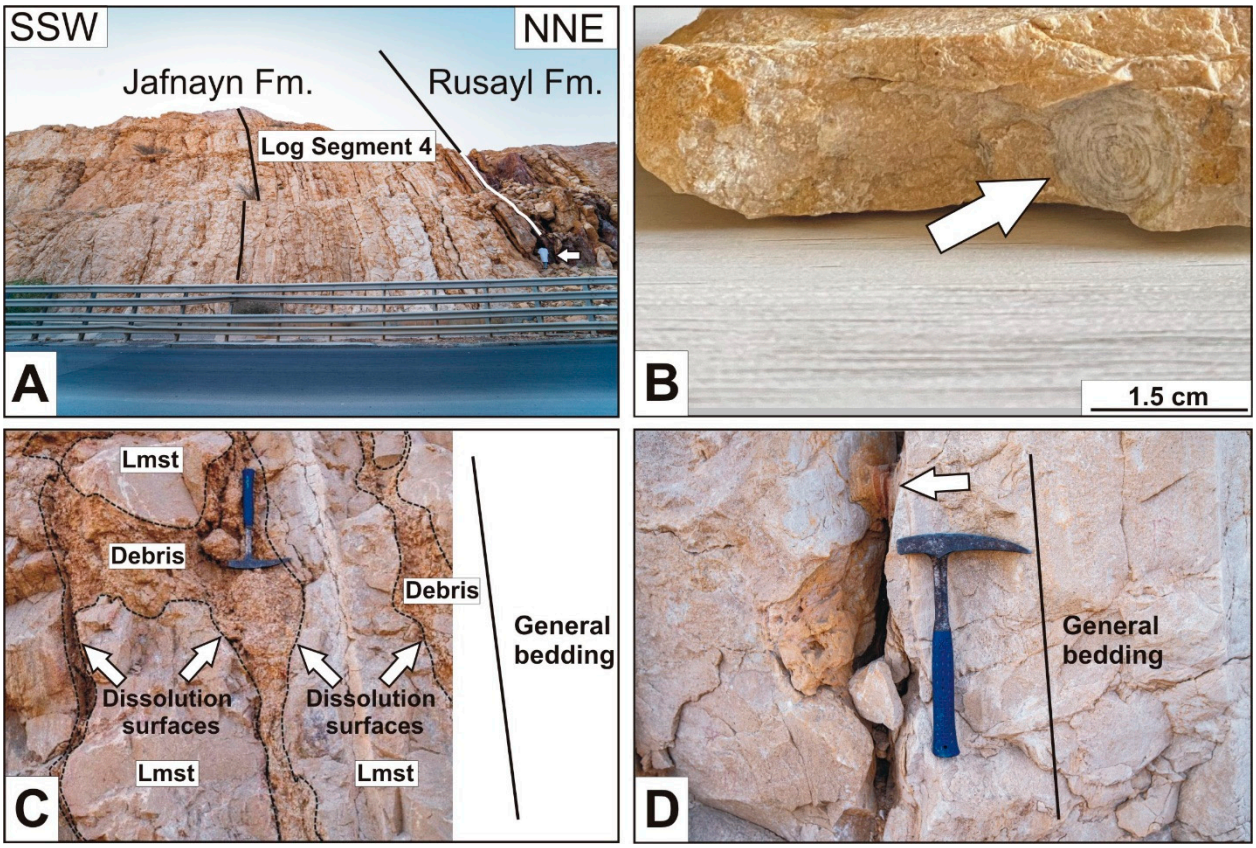


Figure 12. Photographs of Log Segment 4. (**A**) Uppermost part of the Jafnayn Formation and contact to the overlying marl-dominated Rusayl Formation with person for scale marked by arrow. (**B**) Sizeable LBF from 108 m above base. (**C**) Karstification of limestone (light color) at 112 m. Layer-parallel and non-parallel dissolution surfaces (black dotted lines) forming pockets filled with limestone debris (relatively dark). (**D**) Reddish/brownish laminated residual material from karstification (arrow), like iron compounds and clay material. The residual material is less resistant to erosion than the limestone. Lamination is approximately parallel to the general bedding at 116 m.

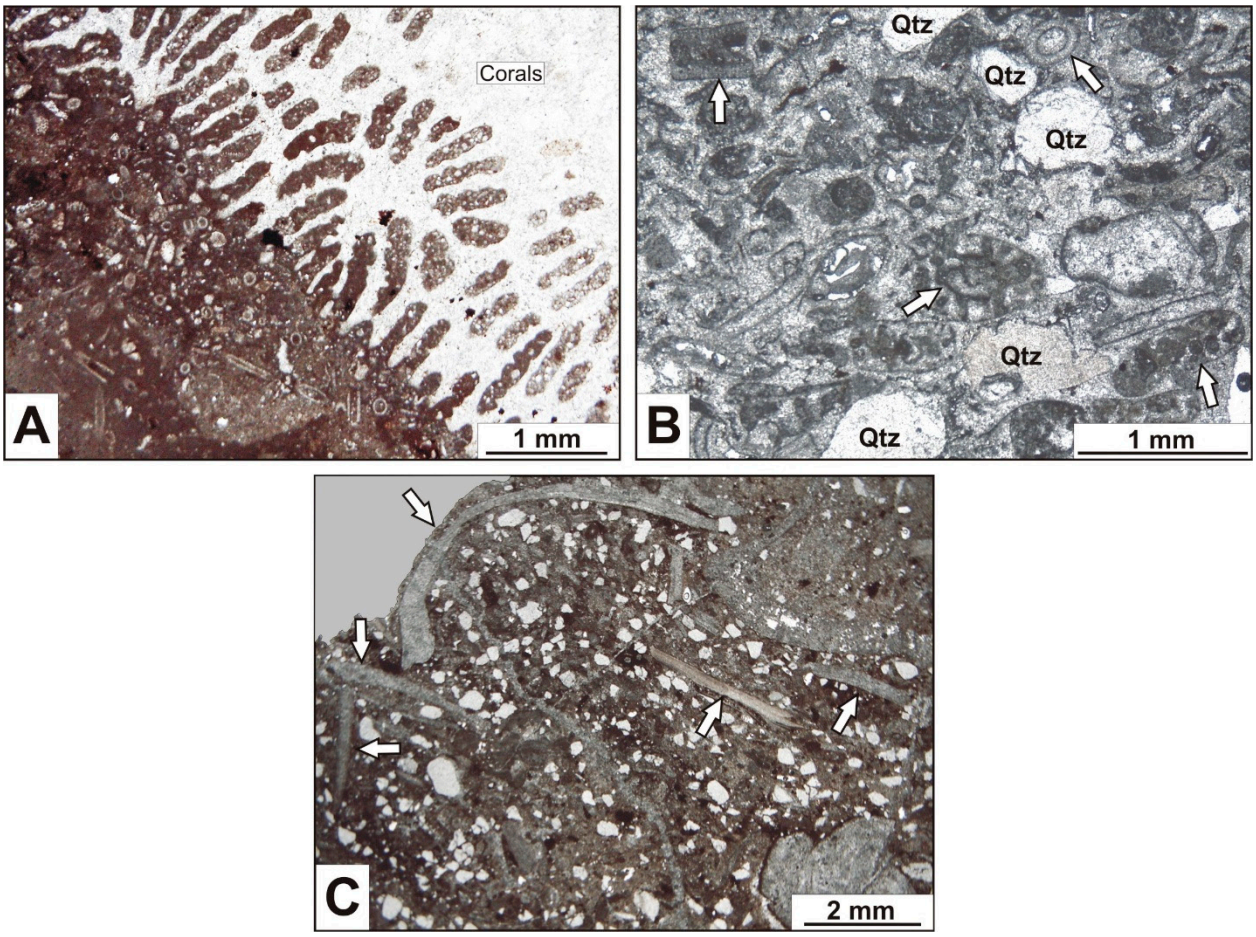


Figure 13. Thin-section photomicrographs of samples from Log Segment 4. (**A**) Corals are the most frequent bioclasts of a coral–red algae floatstone, representing SMF 8. The bioclasts in the micrite are mainly dasycladaceans in different sections. Sample JF2, PPL. (**B**) Quartz arenitic limestone with dasycladaceans (arrows) is the most frequent bioclast type. Most of the detrital quartz debris is marked as "Qtz". Sample JF11, PPL. (**C**) Quartz arenitic limestone with bivalves (arrows) is the most common bioclast. The white components are detrital quartz grains. Sample JF12, PPL.

4.5. Foraminifera

The early Eocene witnessed a global proliferation of LBF species, a process known as the Larger Foraminifera Turnover (LFT), during the Paleocene Eocene Thermal Maximum [48], allowing genera like large *Alveolina*, *Nummulites,* and *Orbitolites* with broad species complexes to flourish as the dominant LBF [48,51]. Formaminiferal taxa of the Jafnayn Formation have been identified [3–7,52].

Tables A1–A4 in the Appendices A–D list the identified foraminiferal taxa for each sample. While Table 1 shows the identified taxa in the different log segments, Figure 14 shows the positions of selected foraminifera in Log Segment 1, which is important, as Log Segment 1 shows the oldest exposed beds.

Benthic foraminifera, which may extend from the Paleocene to the Eocene, were encountered in the Eocene of the studied Jafnayn Formation: *Alveolina* cf. *bronneri* [53], *Alveolina* cf. *elliptica* [54], *Alveolina* cf. *globosa* [6], *Alveolina* cf. *leupoldi* [6], *Alveolina* cf. *oblonga* [55], *Alveolina* cf. *subpyrenaica* [56], *Assilina* sp., *Cibrobulimina* sp., *Discocyclina* sp., *Glomalveolina* sp. [57], *Glomalveolina telemetensis* [58], *Idalina* sp., *Lockhartia prehaimei* [16], *Macetadiscus omanensis*, milioloids, *Nummulites globulus* [59], *Orbitolites* sp., *Pararotalia* sp., *Pyrgo* sp., *Rotalia* sp., *Sakesaria cotteri* [5], *Spirolina* sp., *Thalmanita* sp., *Textularia* sp., and *Triloculina* sp. (Figure 14)

An early Eocene (Ypresian) age of the studied section and a shallow marine depositional environment are indicated by five species of *Alveolina* (e.g., [60]). In addition, these

species represent the upper part of the Jafnayn Formation. These five species are *Alveolina* cf. *globosa* Leymerie, 1846, *Alveolina* cf. *subpyrenaica* Leymerie, 1846, *Alveolina* cf. *leupoldi* Hottinger, 1960, *Alveolina* cf. *oblonga* D'Orbigny, 1826, and *Alveolina* cf. *elliptica* Sowerby, 1840 and are shown in Figure 15A–E. *Alveolina* cf. *elliptica* may also occur in the mid-Eocene [4]. It is of interest that *Alveolina* cf. *elliptica* is present in the lowermost layer of the studied outcrop, not showing any signs of reworking/abrasion (Figure 15A).

Evidence for the late Paleocene (*Sakesaria migiurtina* sp., *Sakesaria nodulifera* sp.) as described by Al-Sayigh [16], was not encountered. The corresponding beds are no longer exposed in the studied section, which is attributed to recent blanketing by scree.

Table 1. Identified foraminifera in the different log segments in alphabetical order.

Log Segment 1	Log Segment 2	Log Segment 3	Log Segment 4
	Alveolina cf. *bronneri*		
Alveolina cf. *elliptica*			
Alveolina cf. *globosa*	*Alveolina* cf. *globosa*		
Alveolina cf. *leupoldi*			
Alveolina cf. *oblonga*			
		Alveolina sp.	
Alveolina cf. *subpyrenaica*			
Assilina sp.	*Assilina* sp.		
Cribrobulimina sp.		*Cribrobulimina* sp.	
		Discocyclina sp.	
Glomalveolina sp.			
Glomalveolina telemetensis			
	Idalina sp.		*Idalina* sp.
	Lockhartia prehaimei		*Lockhartia prehaimei*
Lockhartia sp.			
Macetadiscus omanensis	*Macetadiscus omanensis*		
Miliolid		Miliolid	Miliolid
	Nummulites sp.		
Nummulites globulus			
Orbitolites sp.	*Orbitolites* sp.		*Orbitolites* sp.
			Pararotalia sp.
	Pyrgo sp.		
Rotalia sp.			
			Rotaliids
Sakesaria cotteri	*Sakesaria cotteri*		
			Sakesaria sp.
		Spirolina sp.	
			Thalmanita sp.
Textularia sp.		*Textularia* sp.	*Textularia* sp.
		Triloculina sp.	

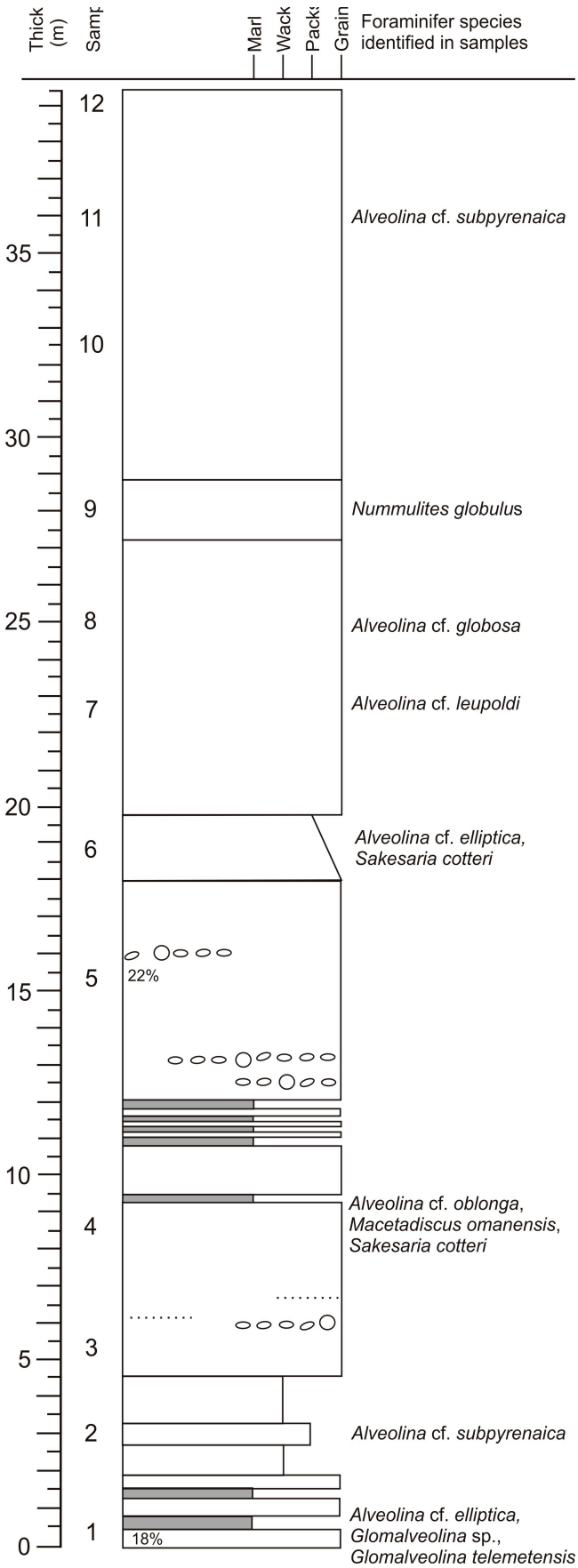


Figure 14. Log Segment 1 (oldest exposed beds) with foraminifer species identified in the respective samples.

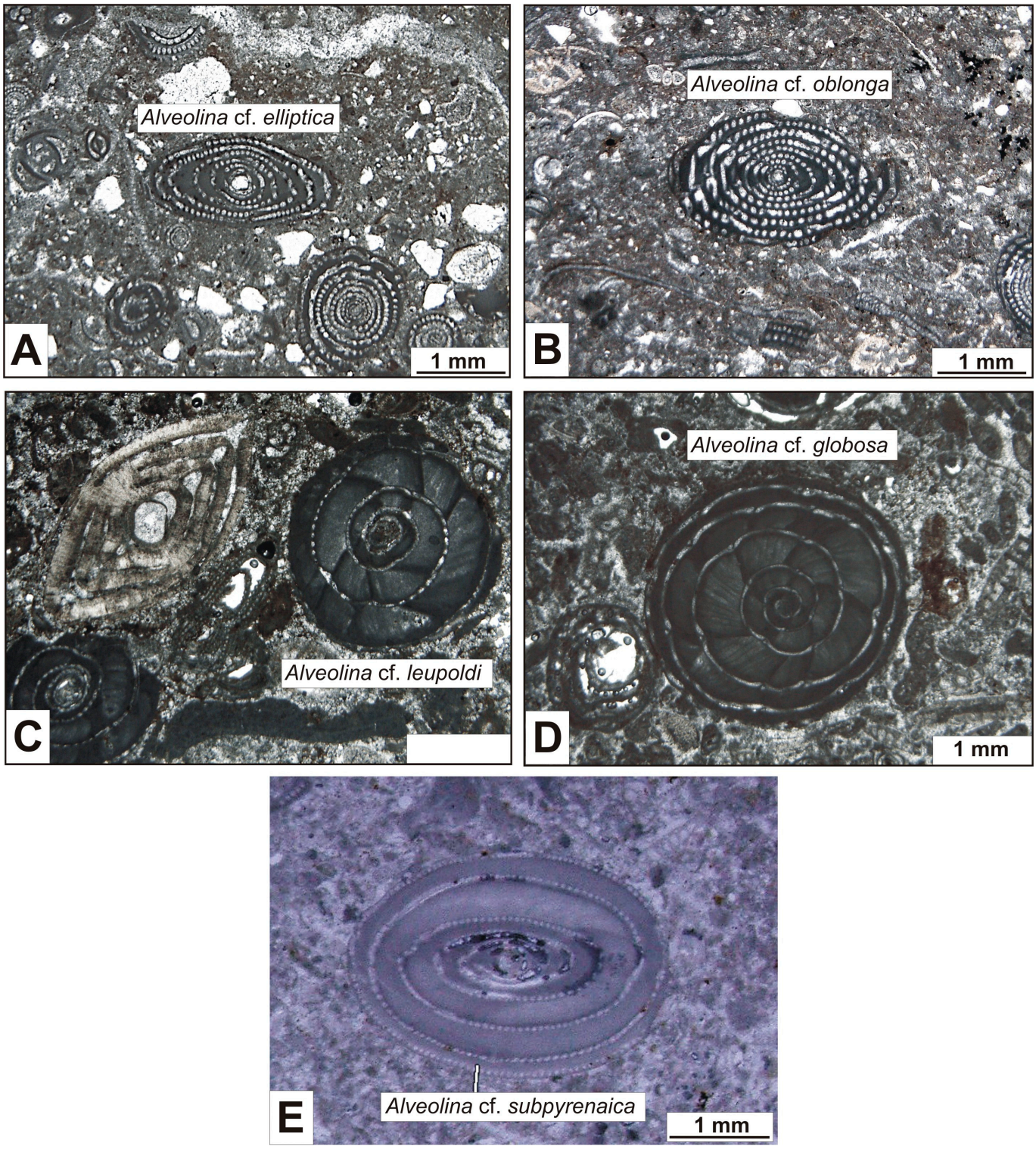


Figure 15. Thin-section photomicrographs of post-Paleocene species of *Alveolina* from the studied section. (**A**–**D**) are PPL; E is PL. (**A**) *Alveolina* cf. *elliptica* (Ypresian and mid-Eocene) from Log Segment 1, Sample 1. (**B**) *Alveolina* cf. *oblonga* (Ypresian) from Log Segment 1, Sample 4. (**C**) *Alveolina* cf. *leupoldi* (Ypresian) from Log Segment 1, Sample 7. (**D**) *Alveolina* cf. *globosa* (Ypresian) from Log Segment 1, Sample 8. (**E**) *Alveolina* cf. *subpyrenaica* (Ypresian) from Log Segment 1, Sample 11.

5. Discussion

In Figure 16, we break down the lithostratigraphy by subdividing it into four different units based on bedding style, limestone types, presence of a few thin to medium marl deposits, purity of limestones, presence of detrital quartz, and prominence/diversity of

bioclasts. Figure 16 also indicates the dominant bioclasts and the depositional environments.

As the basal contact with the underlying Al-Khod Formation is not exposed, and as we found early Eocene foraminifera in the basal bed of Log Segment 1 (Sample 1) without any signs of reworking, the Paleocene and the PETM are not represented in the studied western roadside outcrop. This finding is in agreement with the fact that meters-thick yellow marl, which has been described from the lower part of the formation (Section 1) by Nolan [1] for the Jafnayn area and which is also exposed 8.5 km SSW of Jafnayn (23°26′37.03″ N/58°10′40.31″ E), is missing. This marl represents the lower part of the Jafnayn Formation and is likely the absent late Paleocene part of the formation in the Al-Khod road cut.

We recognized four SMF types: SMF 18-FOR (Figure 6B), SMF 16 (Figure 6B), SMF 18-DASY (Figure 10D), and SMF 8 (Figure 10A–C), the most common of which is SMF 18-FOR. In general, SMF 18 may indicate a restricted (FZ 8) or an open (FZ 7) marine environment [15]. Because SMF 18-FOR is associated with the common SMF 16 (especially in the lower part of the section; Figures 1 and 2), which represents FZ 8, we suggest that SMF 18 also indicates FZ 8. Facies zone 8 denotes a restricted lagoonal environment with water depths reaching a few tens of meters within the euphotic zone [15]. Thus, the formation predominantly represents a restricted lagoonal environment (Figures 4, 7, 9, 11, and 16).

The peloids in the peloidal grainstones and packstones of SMF 16 signify sites of reduced sedimentation rates [61] in comparison to those of SMF 18. As all ostracods are smooth-shelled, low-energy conditions are indicated for various levels of the studied section.

The occurrence of SMF 8 at three stratigraphic levels and the single occurrence of SMF 18-DASY indicate FZ 7 (Figure 9), which implies open marine conditions with a water depth of a few meters to tens of meters within the euphotic zone [15]. FZ 7 is normally above fair-weather wave base, but SMF 8 occurs below fair-weather wave base in shelf lagoons with circulation in low-energy environments [15]. The stratigraphically limited occurrences of SMF 8 and SMF 18-DASY (Figures 9 and 11) indicate brief intervals of open marine conditions within a generally restricted lagoon (Figures 9 and 11). Facies zone 7 was observed in the upper part of the section, indicating that the water exchange with the open ocean improved towards the top (Figure 16).

In the upper part of the studied section, SMF 8 and SMF 18-DASY are associated with SMF 18-FOR. Therefore, it is possible that SMF 18-FOR may represent FZ 7 in the upper part of the section (see above). In this case, open marine conditions in the upper part of the studied section could be more widely distributed, as shown in Figure 16. Increasing open marine conditions towards the top may also be reflected by an increase in coral debris in the same direction (Figures 4, 7, 9, 11, and 16). However, the multi-faceted research approach by Dill et al. [12] indicated a shallow lagoon with moderately reducing conditions for the uppermost part of the studied section, favoring the idea of a restricted environment. All ostracods, including those of log segments 3 and 4, indicate low-energy conditions, which goes along better with restricted than with open marine conditions. In addition, the peloids in the peloidal grainstones and packstones of SMF 16 (log segments 3 and 4) and their indication of reduced sedimentation rates [61] also tend to favor restricted over open marine conditions. Thus, we prefer the interpretation that the upper part of the section was also dominated by restricted environments.

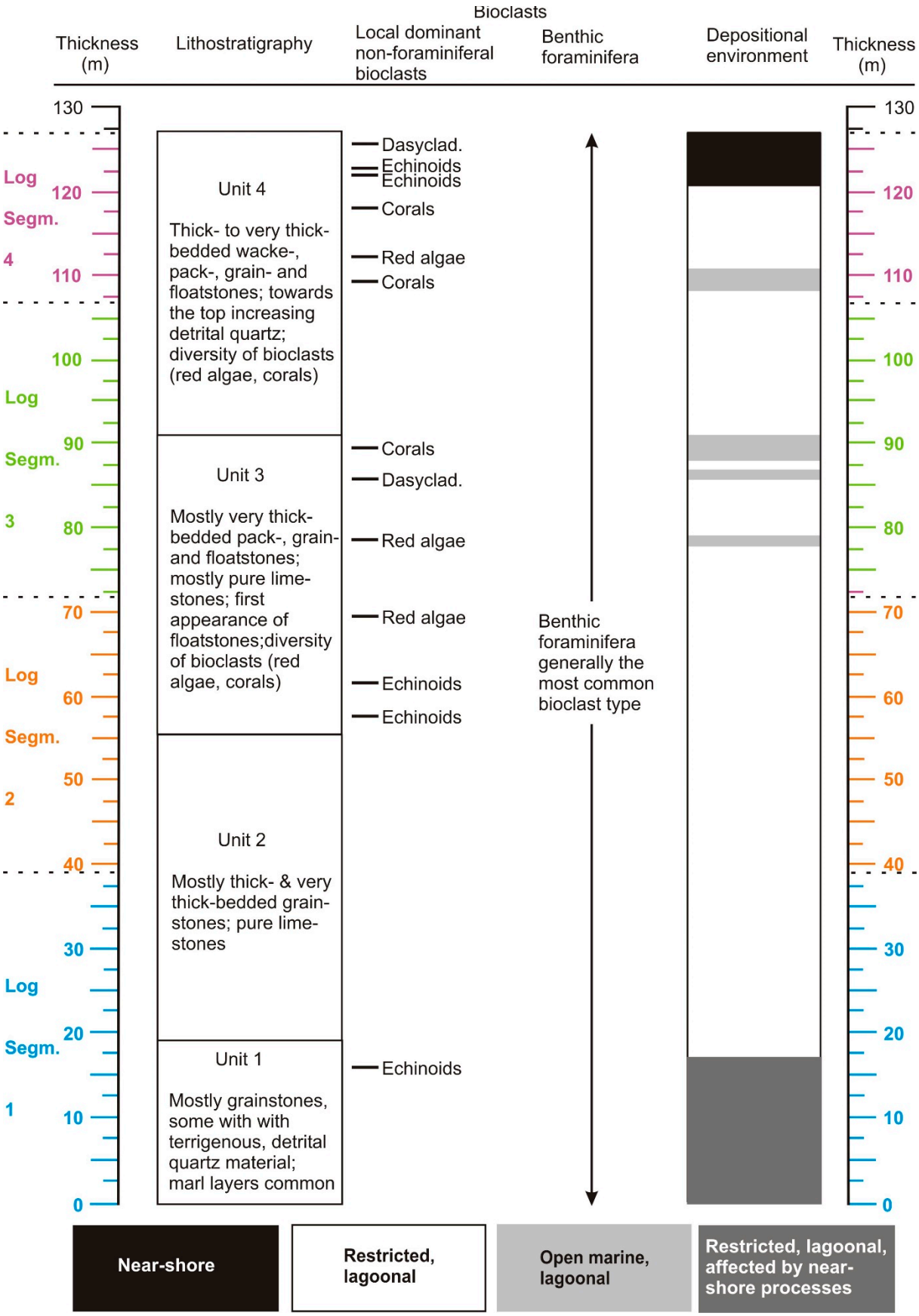


Figure 16. Lithostratigraphic organization, dominant bioclasts, and depositional environments of the studied section of the Jafnayn Formation. Open marine environments are more common in the

upper part of the section. They could be more frequent in the upper part than shown (see text for discussion).

Terrigenous influx is generally common in FZ 8 [15] and is especially noticeable in the lower part (0–~17 m) and the upper part (~117–127 m) of the studied outcrop, seemingly reflecting a paleogeographic change at the transition to the overlying Rusayl Formation, which is characterized by marl, limestone, and sandstone. Detrital quartz within the basal and uppermost parts of the Jafnayn Formation has been observed at other locations by Nolan et al. [1], Racey [4], and Tomás et al. [7]. This holds true for the Sunub area as well. Since the Paleocene is not represented in the studied section, we note that coarse material may also occur in the middle part of the Jafnayn Formation.

The distribution of the dominant SMFs and FZs detailed above indicates that the basal 17 m of the Jafnayn Formation consist of lagoonal deposits with periodic influx of detrital quartz material, including conglomerate horizons with subrounded clasts and angular to well-rounded quartz sand. We suggest that this part of the lagoon was relatively close to shore, in the reach of estuarine sediment influx at times of high discharge events, during which coarse siliciclastic detritus was transported into the lagoon.

The uppermost 6–7 m of the Jafnayn Formation lack SMF assignments (Figure 11) and are interpreted as near-shore deposits based on the common occurrence of quartz grains, reflecting a terrestrial influence. In line with this is their textural maturity range from angular to well-rounded. While well-rounded quartz grains may indicate reworking by long-shore currents in a beach/near-beach setting, a nearby estuary may have led to the influx of angular detrital quartz grains. The angular to well-rounded grains may have been mixed or homogenized by long-shore, tidal, and rip currents.

The lagoon interpretation raises the question as to what kind of material may have formed the barrier(s) to create the protected environment. Considering the identified allochems and their quantities, the abundance of foraminiferal tests could suggest that shoals or barrier beaches were largely made of foraminiferal remains (in line with 7) and other skeletal and non-skeletal grains that were reworked from the lagoon to form shoals. For the upper part of the formation, where corals are common, coral build-ups may be considered the main structures to protect the lagoon. The presence of coral banks was recognized for the Paleocene to Eocene of the Arabian platform [62]. It should be noted that corals were not observed in the Jafnayn Formation 150 km to the southeast in a detailed study [7].

Possible causes for the development of the brief open marine intervals are sea-level rises above the lagoon barriers, allowing for better water exchange with the open sea. Short-term eustatic sea-level rises may be considered. Episodes of regional gravitational collapse along the Frontal Range Fault could have caused subsidence to the north of the fault on the hanging-wall block, considering that the fault was active during deposition of the Jafnayn Formation [10]. These movements would have caused relative sea-level rises. These rises must have counteracted the effects of doming of the Jabal Akhdar and Saih Hatat areas [43,63], which are expected to have generally favored a drop of the relative sea-level.

The influx of detrital quartz and the conglomerate horizons and their associations with the shallow marine limestones may indicate the proximity of the Oman Mountains, from where this material was shed.

The absence of late Paleocene foraminifera suggests that the PETM event is not represented in the exposed section. Thus, the measured logs must have been deposited during the overall cooling of the Eocene [46,49].

6. Conclusions

The lithostratigraphic and facies findings presented in this paper are, overall, compatible with those of Nolan et al. [1], Racey [4], and Tomás et al. [7]. Considering the

absence of exposed Paleocene beds in the studied section, the Jafnayn Formation must be thicker than at the type section in the western Rusayl Embayment.

The formation at the studied site mainly accumulated in a restricted lagoon, interrupted by intervals of open marine conditions caused by short-term eustatic sea-level rises and/or regional movements due to gravitational collapse.

The dominating bioclasts are benthic foraminifera, which may have been significant contributors to the formation of the lagoon barrier during specific time intervals. During deposition of the upper part of the sampled Jafnayn Formation, coral banks may have protected the lagoon (uppermost part of Unit 2 and units 3 and 4).

The lower and upper parts of the formation display noticeable quartz detritus, which appears to be typical for the formation, not only at the studied site. The detrital quartz material near the base and top coincides (1) with a restricted lagoonal environment affected by near-shore processes (base; Unit 1) and (2) with a near-shore environment where SMFs cannot be assigned (top, Unit 4). While doming of the Jabal Akhdar and Saih Hatat areas [43,63] must have favored a drop of the relative sea-level, episodes of rising sea-levels may reflect gravitational collapse along the Frontal Range Fault and subsidence on the northerly located hanging-wall block.

Open marine conditions coincide with diverse bioclastic fragments of the formation (units 3 and 4; Figure 16). There is a possibility that a few more open marine deposits exist that were left unsampled and, thus, unnoticed.

The detrital quartz and the conglomerate horizons among the limestones of the Jafnayn Formation reflect the proximity of the Oman Mountains, which served as a source for these materials.

The absence of late Paleocene beds suggests that the PETM event is not represented in the exposed section; it must have been deposited during the overall cooling of the Eocene.

Regarding future research related to biostratigraphy, chemostratigraphy (e.g., solving paleoclimate questions), and regional paleofacies mapping, our results may rule in or rule out the famous Al-Khod roadcut for consideration of such studies.

Author Contributions: Conceptualization: F.M.; formal analysis: F.M., A.S., A.R.A.-S., A.A.-M., L.G., S.A.-G., G.F., L.C., M.A.-S., and F.A.-J.; methodology: F.M., A.R.A.-S., L.G., G.F., and L.C. investigation: F.M., A.S., A.R.A.-S., A.A.-M., L.G., S.A.-G., G.F., L.C., M.A.-S., and F.A.-J.; writing—original draft preparation: F.M.; writing—review and editing: F.M., A.S., A.R.A.-S., L.G., G.F., and L.C.; visualization: F.M. and A.S.; supervision: F.M. and A.R.A.-S. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. Summary of microscopic sample information of the basal 39.5 m of the studied section from the base (Sample 1) to the top (Sample 12). The relative frequency of bioclasts and other allochems is indicated. Limestones with 10–50% are not considered true limestones, but "quartz arenititic limestones". Their SMF/FZ assignments are kept in parentheses.

Sample	Rock Classification	Contents of Bioclasts, Other Allochems, Detrital Quartz	SMF/FZ
12 (Top)	Locally dolomitized foraminiferal grainstone	Benthic foraminifera, including *Alveolina* sp., possibly *Alveolina* cf. *elliptica* > echinoids, including spines > ostracods Common peloids Dolomite crystals have dark cores and limpid rims, are rhombic and non-mimetic; ≤0.15 mm	SMF 18-FOR FZ 8
11	Widely dolomitized foraminiferal grainstone	Benthic foraminifera, including *Alveolina* cf. *subpyrenaica* > echinoids > ostracods = branching red algae Dolomite crystals are limpid, rhombic, and non-mimetic; ≤0.1 mm	SMF 18-FOR FZ 8
10	Fine-grained peloidal grainstone	Benthic foraminifera, including miliolids, *Nummulites* sp. > echinoids, including spines and corona plates > ostracods > dasycladaceans Abundant peloids	SMF 16-NON-LAMINATED FZ 8
9	Foraminiferal grainstone	Benthic foraminifera, including *Alveolina* sp., miliolids, *Nummulites globulus* > echinoids, including spines > ostracods Few peloids	SMF 18-FOR FZ 8
8	Foraminiferal grainstone	Benthic foraminifera, including *Alveolina* cf. *globosa*, *Alveolina* sp. (Ypresian), *Nummulites* sp. > echinoids Few peloids	SMF 18-FOR FZ 8
7	Locally dolomitized foraminiferal grainstone	Benthic foraminifera, *Alveolina* cf. *leupoldi*, *Alveolina* sp. (Ypresian) > echinoids, including spines > bivalves Few peloids Dolomite crystals are limpid, rhombic, and non-mimetic; ≤0.1 mm	SMF 18-FOR FZ 8
6	Foraminiferal pack- to grainstone	Benthic foraminifera, including *Alveolina* cf. *elliptica*, *Assilina* sp., *Sakesaria cotteri*, *Textularia* sp. > echinoids, including spines and corona plates > ostracods 8% monocrystalline subangular to well-rounded fine to coarse quartz sand; one well-rounded crystalline dolomite clast of 1.3 mm diameter	SMF 18-FOR FZ 8
5	Laminated echinoid grainstone, if detrital quartz content of >10% is not considered; quartz arenitic limestone (no true limestone)	Echinoids, including spines > benthic foraminifera, including *Rotalia* sp. 22% monocrystalline angular to well-rounded fine to coarse quartz sand	-
4	Foraminiferal grainstone	Benthic foraminifera, including *Alveolina* cf. *oblonga*, *Alveolina* sp., *Sakesaria cotteri*, *Sakesaria* sp., *Macetadiscus omanensis*, *Orbitolites* sp. > ostracods > bivalves 4% monocrystalline angular to well-rounded fine to coarse quartz sand	SMF 18-FOR FZ 8
3	Fine-grained peloidal grainstone	Benthic foraminifera, including miliolids, *Nummulites* sp. > ostracods > echinoids, including spines Abundant peloids	SMF 16-NON-LAMINATED FZ 8

		3% monocrystalline angular very fine to fine quartz sand	
2	Foraminiferal packstone	Benthic foraminifera, including *Alveolina* cf. *subpyrenaica*, *Cribrobulimina* sp. (Paleocene-Eocene), *Glomalveolina* sp., *Lockhartia* sp., *Textularia* sp. > dasycladaceans > ostracods > echinoids, including spines	SMF 18-FOR FZ 8
1 (Base)	Foraminifer packstone, if detrital quartz content of >10% is not considered; quartz arenitic limestone (no true limestone)	Benthic foraminifera, including *Alveolina* cf. *elliptica*, *Glomalveolina* sp., *Glomalveolina telemetensis*, miliolids > ostracods > dasycladaceans 18% of angular to well-rounded, fine to very coarse quartz sand; quartz grains mainly monocrystalline	(SMF 18-FOR FZ 8)

Appendix B

Table A2. Summary of microscopic sample information from 39.5 to 72 m above base of the studied section from the base (Sample S1) to the top (Sample S12). The relative frequency of bioclasts and other allochems is indicated. Order of sample numbering: Note that the positions of samples S7 to S10 are as shown on the table!

Sample	Rock Classification	Contents of Bioclasts, Other Allochems, Detrital Quartz	SMF/FZ
S12 (Top)	Foraminiferal grainstone	Benthic foraminifera, including *Sakesaria cotteri*, *Alveolina* sp. > echinoids, including spines and corona plates > poritid corals > branching and encrusting red algae > bivalves = ostracods > dasycladaceans Few peloids, few cortoids	SMF 18-FOR FZ 8
S11	Foraminiferal packstone	Benthic foraminifera, including *Alveolina* cf. *globosa*, *Orbitolites* sp. > echinoids, including spines > red algae; encrusting types, branching ones and nodules (rhodoids) = poritid corals> ostracods Few peloids	SMF 18-FOR FZ 8
S8	Wackestone	Red algae; encrusting types (encrusting corals), nodular types (rhodoids) and branching ones > benthic foraminifera, including *Idalina* sp., *Lockhartia prehaimei*, *Lockhartia* sp., > poritid corals > echinoids, including spines > ostracods	-
S7	Dolostone	A few preserved echinoid fragments, including a spine > foraminifer *Pyrgo* sp. and a few ghosts of benthic foraminifera Dolomite crystals are mostly dark, anhedral, and non-mimetic (≤0.2 mm)	-
S10	Locally dolomitized foraminiferal packstone	Benthic foraminifera, including *Alveolina* sp., *Nummulites* sp. > echinoids, including spines > bivalves > branching (rhodophytic) red algae > ostracods Few peloids Dolomite crystals have dark cores and limpid rims, are rhombic and non-mimetic; ≤0.1 mm	SMF 18-FOR FZ 8
S9	Peloidal grainstone	Echinoids, including spines > benthic foraminifera, including *Macetadiscus omanensis* > poritid corals > red algae Abundant peloids	SMF 16-NON-LAMINATED FZ 8
S6	Widely dolomitized foraminiferal limestone (wacke- to packstone?)	Benthic foraminifera, including *Alveolina* cf. *globosa*, *Nummulites* sp. Dolomite crystals have dark cores and limpid rims and are rhombic and non-mimetic (≤0.15 mm) or limpid, rhombic, and non-mimetic (≤0.1 mm)	-

S5	Foraminiferal packstone	Benthic foraminifera > echinoids, including spines Few peloids	SMF 18-FOR FZ 8
S4	Foraminiferal grainstone	Benthic foraminifera, including *Alveolina* cf. *bronneri*, *Idalina* sp., > gastropods > dasycladaceans > echinoids	SMF 18-FOR FZ 8
S3	Foraminiferal packstone	Benthic foraminifera, including globular *Alveolina* sp., *Nummulites* sp. > echinoids, including spines > ostracods = bivalves = green algae, including dasycladaceans > gastropods	SMF 18-FOR FZ 8
S2	Foraminiferal grainstone	Benthic foraminifera, including *Alveolina moussoulensis*?, miliolids > echinoids, including corona plates and spines > red algae, including *Distychoplax biserialis* > bivalves = gastropods = dasycladaceans = ostracods = bryozoans	SMF 18-FOR FZ 8
S1 (Base)	Locally dolomitized foraminiferal grainstone	Benthic foraminifera, including *Alveolina* sp., *Assilina* sp. > echinoids > red algae Some peloids Dolomite crystals have dark cores and limpid rims and are rhombic and non-mimetic; ≤0.15 mm	SMF 18-FOR FZ 8

Appendix C

Table A3. Summary of microscopic sample information from 72 to 107 m above base of the studied section from the base (Sample 1.) to the top (Sample 12.). The relative frequency of bioclasts and other allochems is indicated. Please note that dots after the numbers are part of the sample code to avoid confusion with the sample codes of Log Segment 1!

Sample	Rock Classification	Contents of Bioclasts, Other Allochems, Detrital Quartz	SMF/FZ
12. (Top)	Peloidal grainstone	Benthic foraminifera, including miliolids, *Triloculina* sp., *Spirolina* sp., *Textularia* sp. = echinoids, including spines > dasycladaceans > bivalves > ostracods Abundant peloids 1% monocrystalline angular fine-grained quartz sand	SMF 16-NON-LAMINATED FZ 8
11.	Foraminiferal grainstone	Benthic foraminifera, including *Cribrobulimina* sp.? > poritid corals > encrusting red algae > echinoids, including spines > dasycladaceans > ostracods Some peloids, few cortoids	SMF 18-FOR FZ 8
10.	Locally dolomitized coral-red algae floatstone	Poritid corals > encrusting red algae (encrusting corals PHOTO) > benthic foraminifera > ostracods = gastropods = dasycladaceans > echinoids Dolomite crystals have dark cores and limpid rims, are rhombic and non-mimetic; ≤0.15 mm	SMF 8 FZ 7
9.	Peloidal grainstone	Benthic foraminifera, including *Triloculina* sp. > echinoids, including spines > dasycladaceans > ostracods > bivalves Abundant peloids, some cortoids 4% monocrystalline angular to subangular fine to medium quartz sand	SMF 16-NON-LAMINATED FZ 8
8.	Dasycladacean grainstone	Dasycladaceans > benthic foraminifera, including miliolids, *Triloculina* sp. > nodular red algae (rhodoids) > bivalves > echinoids, including spines > ostracods Some peloids, some cortoids	SMF 18-DASY FZ 7
7.	Foraminferal grainstone	Benthic foraminifera, including miliolids > poritid corals > dasycladaceans > gastropods > echinoids > bivalves > ostracods	SMF 18-FOR FZ 8

6.	Foraminifer grainstone	Benthic foraminifera, including *Cribrobulimina* sp., *Discocyclina* sp. (Thanetian to early Eocene) > echinoids, including numerous sizeable spines > poritid corals > dasycladaceans > ostracods Few peloids	SMF 18-FOR FZ 8
5.	Locally dolomitized peloidal grainstone	Benthic foraminifera, including *Triloculina* sp. > echinoids = bivalves > ostracods Abundant peloids, some micritic miliolid-bearing intraclasts Dolomite crystals are limpid (clear, colorless) or beige, anhedral and non-mimetic; ≤0.15 mm	SMF 16-NON-LAMINATED FZ 8
4.	Coral-red algae floatstone	Red algae; encrusting ones (encrusting corals) and nodular types (rhodoids) > poritid corals > bivalves > benthic foraminifera, including *Alveolina* sp. > echinoids > gastropods Some peloids	SMF 8 FZ 7
3.	Peloidal grainstone	Benthic foraminifera > poritid corals > encrusting, corallinacean red algae > echinoids, including spines > dasycladacean green algae (group group *Thyrsoporelleae)* > bivalves > ostracods. Abundant peloids 3% monocrystalline angular to subangular fine quartz sand	SMF 16-NON-LAMINATED FZ 8
2.	Dolostone	Few preserved benthic foraminifera, including miliolids > echinoids > ostracods	-
1. (Base)	Locally dolomitized intraclastic peloidal grainstone	Benthic foraminifera > echinoids, including spines > gastropods > ostracods Intraclasts and abundant peloids Dolomite crystals are limpid, rhombohedral, and non-mimetic; ≤0.15 mm	-

Appendix D

Table A4. Summary of microscopic sample information of the uppermost 20 m of the studied section from the base (Sample JF1) to the top (Sample JF12). The relative frequency of bioclasts and other allochems is indicated. Limestones with 10–50% are not considered true limestones, but "quartz arenititic limestones". Order of sample numbering: Note that the positions of samples JF10 and JF11 are as shown on the table!

Sample	Rock Classification	Contents of Bioclasts, Other Allochems, Detrital Quartz	SMF/FZ
JF12 (Top)	Locally dolomitized wacke- to floatstone if detrital quartz content of >10% is not considered; quartz arenitic limestone (no true limestone)	Bivalves > benthic foraminifera, including *Pararotalia* sp., *Sakesaria* sp. > echinoids, including spines > ostracods Few cortoids 28% monocrystalline angular to subrounded fine to coarse quartz sand Dolomite crystals are beige and anhedral; ≤0.1 mm	-
JF10	Quartz arenitic limestone	Benthic foraminifers, including *Idalina* sp., *Textularia* sp. Fine-grained calciclastic and siliciclastic components 40% monocrystalline angular fine quartz sand	-
JF11	Grainstone if detrital quartz of >10% is not considered; quartz arenitic limestone (no true limestone)	Dasycladaceans > benthic foraminifera, including *Alveolina* sp., *Orbitolites* sp., *Textularia* sp. > gastropods > echinoids, including spines Some cortoids 20% mostly monocrystalline angular to very well-rounded fine to very coarse quartz sand	-

JF9	Wackestone if detrital quartz if content of >10% is not considered; quartz arenitic limestone (no true limestone)	Echinoids, including spines > benthic foraminifera, including *Alveolina* sp., *Lockhartia prehaimei*, miliolids, *Orbitolites* sp. > ostracods > corals 40% mostly monocrystalline angular to very well-rounded fine to coarse quartz sand and some quartz granules	-
JF8	Packstone if detrital quartz if content of >10% is not considered; quartz arenitic limestone (no true limestone)	Echinoids > benthic foraminifera, including *Alveolina* sp., rotaliids 30% monocrystalline angular to subangular fine quartz sand	-
JF7	Foraminiferal packstone	Benthic foraminifera > echinoids, including spines > bivalves > ostracods Few peloids	SMF 18-FOR FZ 8
JF6	Peloidal packstone	Poritid corals > benthic foraminifera > echinoids, including spines > dasycladaceans > ostracods Abundant peloids 1% mono- and polycrystalline well-rounded coarse quartz sand	SMF 16-NON-LAMINATED FZ 8
JF5	Partly dolomitized foraminiferal grainstone	Benthic foraminifera > echinoids > dasycladaceans 8% of monocrystalline angular to subangular fine quartz sand Dolomite crystals are beige, rhombic and non-mimetic; ≤0.15 mm	SMF 18-FOR FZ 8
JF4	Foraminiferal grainstone	Benthic foraminifera, including *Alveolina* sp. (Ypresian) > echinoids > bivalves > gastropods Frequent peloids, some cortoids	SMF 18-FOR FZ 8
JF3	Locally dolomitized wackestone	Encrusting red algae > benthic foraminifera, including miliolids > bivalves > echinoids > gastropods Some peloids 1% angular to subrounded fine to medium monocrystalline quartz sand Dolomite crystals are dark, anhedral, and non-mimetic; ≤0.1 mm	-
JF2	Locally dolomitized coral-red algae floatstone	Corals > red algae, encrusting corals > bivalves > benthic foraminifera, including *Idalina* sp., miliolids, *Textularia* sp. > echinoids = ostracods > dasycladaceans > gastropods Few peloids Dolomite crystals are beige, anhedral, and non-mimetic; ≤0.2 mm	SMF 8 FZ 7
JF1 (Base)	Peloidal grainstone	Benthic foraminifera, including *Thalmanita* sp. > echinoids, including spines and corona plates > ostracods > bivalves Abundant peloids, few cortoids 8% angular to subrounded fine to medium monocrystalline quartz sand	SMF 16-LAMINATED FZ 8

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